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Engineered human-endogenous virus-like particles and methods of use thereof for delivery to cells

Abstract

Human-derived virus-like particles (heVLPs), comprising a membrane comprising a phospholipid bilayer with one or more HERV-derived envelope proteins on the external side; one or more HERV-derived GAG proteins in the heVLP core, and a cargo molecule, e.g., a biomolecule and/or chemical cargo molecule, disposed in the core of the heVLP on the inside of the membrane, wherein the heVLP does not comprise a gag protein, except for gag proteins that are encoded in the human genome or gag proteins that are encoded by a consensus sequence that is derived from gag proteins found in the human genome, and methods of use thereof for delivery of the cargo molecule to cells.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 17/617,490, filed Dec. 8, 2021, which is a § 371 National Stage Application of PCT/US2020/037740, filed Jun. 15, 2020, which claims the benefit of U.S. Patent Application Ser. No. 62/861,186, filed on Jun. 13, 2019. The entire contents of the foregoing are hereby incorporated by reference.

SEQUENCE LISTING

(1) This application contains a Sequence Listing that has been submitted electronically as an XML file named '29539-0385004_SL_ST26.xml'. The XML file, created on Jul. 10, 2023, is 136,545 bytes in size. The material in the XML filed is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) Described herein are engineered human-endogenous virus-like particles (heVLPs) comprising a membrane comprising a phospholipid bilayer on the external side; and a cargo, e.g., a biomolecule and/or chemical cargo, disposed in the core of the heVLP on the inside of the membrane, wherein the heVLP does not comprise a protein from non-human gag or pol, and methods of use thereof for delivery of the cargo to cells.

BACKGROUND

(3) Delivery of cargo such as proteins, nucleic acids, and/or chemicals into the cytosol of living cells has been a significant hurdle in the development of biological therapeutics.

SUMMARY

(4) Described herein are heVLPs that are capable of packaging and delivering DNA, RNA, protein, chemical compounds and/or molecules, and any combination of these four entities into eukaryotic cells. The non-viral heVLP systems described herein have the potential to be simpler, more efficient and safer than conventional, artificially-derived lipid/gold nanoparticles and viral particle-based delivery systems because heVLPs are comprised of human-derived components. The cargo inside may or may not be human derived, but the heVLP is entirely comprised from human and synthetic non-immunogenic components. “Synthetic” components include surface scFv/nanobody/darpin peptides that have been demonstrated to not be immunostimulatory and can be used to enhance targeting and cellular uptake of heVLPs. This means that the exterior surface of the particle lacks components that are significantly immunostimulatory, which should minimize immunogenicity and antibody neutralization of these particles. Excluding cargo, the heVLPs do not contain exogenous viral components inherent to other VLPs and this represents a significant and novel advancement in technology. In addition, heVLPs can utilize (but do not require) chemical-based dimerizers, and heVLPs have the ability to package and deliver cargo molecules including therapeutic or diagnostic agents, including biomolecules and chemicals, e.g., specialty single and/or double-stranded DNA molecules (e.g., plasmid, mini circle, closed-ended linear DNA, AAV DNA, episomes, bacteriophage DNA, homology directed repair templates, etc.), single and/or double-stranded RNA molecules (e.g., single guide RNA, prime editing guide RNA, messenger RNA, transfer RNA, long non-coding RNA, circular RNA, RNA replicon, circular or linear splicing RNA, micro RNA, small interfering RNA, short hairpin RNA, piwi-interacting RNA, toehold switch RNA, RNAs that can be bound by RNA binding proteins, bacteriophage RNA, internal ribosomal entry site containing RNA, etc.), proteins, chemical compounds and/or molecules (e.g., small molecules), and combinations of the above listed cargos (e.g. AAV particles).

(5) The heVLPs described herein are different from conventional retroviral particles, virus-like particles (VLPs), exosomes and other previously described extracellular vesicles that can be loaded with cargo, at least because heVLPs can be produced by a strategic overexpression of human-derived components in human cells, heVLPs have a vast diversity of possible cargos and loading strategies, heVLPs lack a limiting DNA/RNA length constraint, heVLPs lack proteins derived from pol and exogenous gag, and heVLPs have unique mechanisms of cellular entry.

(6) Described herein are compositions and methods for cargo delivery that can be used with a diverse array of protein and nucleic acid molecules, including genome editing, epigenome modulation, transcriptome editing and proteome modulation reagents, that are applicable to many disease therapies.

(7) Thus, provided herein are engineered heVLPs, comprising a membrane comprising a phospholipid bilayer with one or more HERV-derived ENV/glycoprotein(s) (e.g., overexpressed from exogenous sources, such as plasmids or stably integrated transgenes, in heVLP production cells) (e.g., as shown in Table 1) on the external side; and a human endogenous GAG protein, other plasma membrane recruitment domain, and/or biomolecule/chemical cargo disposed in the core of the heVLP on the inside of the membrane, wherein the biomolecule cargo may or may not be fused to a human-endogenous GAG or other plasma membrane recruitment domain (e.g., as shown in Table 6), and the heVLP does not comprise a non-human gag and/or pol protein, do not express gag and/or pol proteins except for gag proteins that are encoded in the human genome or gag proteins that are encoded by a consensus sequence that is derived from gag proteins found in the human genome. Human-derived GAG or other plasma membrane recruitment domains fused to biomolecule cargo can be overexpressed from exogenous sources, such as plasmids or stably integrated transgenes, in heVLP production cells.

(8) In some embodiments, the HERV ENV can be truncated or fused to an scFv or other targeting

polypeptides.

(9) In some embodiments the HERV GAG can be fused to a plasma membrane recruitment domain (e.g., as shown in Table 6).

(10) In another embodiment, engineered heVLPs comprise a membrane comprising a phospholipid bilayer with one or more HERV-derived ENV/glycoprotein(s) (e.g., overexpressed from exogenous sources, such as plasmids or stably integrated transgenes, in heVLP production cells) (e.g., as shown in Table 1) on the external side; and, if desired, a plasma membrane recruitment domain (e.g., as shown in Table 6); and, if desired a biomolecule/chemical cargo inside the particle.

(11) Also provided are methods of delivering a cargo to a target cell, e.g., a cell in vivo or in vitro, by contacting the cell with the heVLP of claim 1 comprising the biomolecule and/or chemical as cargo.

(12) In addition, provided herein are methods for producing a heVLP comprising a biomolecular cargo. The methods include providing a cell expressing (e.g., engineered to express or overexpress) one or more HERV-derived envelope proteins (e.g., as shown in Table 1), and a cargo, wherein the cell does not express a gag and/or pol protein, except for gag proteins that are encoded in the human genome or gag proteins that are encoded by a consensus sequence that is derived from gag proteins found in the human genome; and maintaining the cell under conditions such that the cells produce heVLPs. In some embodiments, the methods further include harvesting and optionally purifying and/or concentrating the produced heVLPs.

(13) Also provided herein are cells (e.g., isolated cells, preferably mammalian, e.g., human, cells) that express, e.g., that have been induced to overexpress, in combination one or more HERV-derived envelope proteins (e.g., (overexpressed from exogenous sources, such as plasmids or stably integrated transgenes)(e.g., as shown in Table 1), and a cargo fused to a human endogenous GAG or other plasma membrane recruitment domain (e.g., as shown in Table 6), wherein the cell does not express a gag protein except for gag proteins that are encoded in the human genome or gag proteins that are encoded by a consensus sequence that is derived from gag proteins found in the human genome (overexpressed from exogenous sources, such as plasmids or stably integrated transgenes)). In some embodiments, the cells are primary or stable human cell lines, e.g., Human Embryonic Kidney (HEK) 293 cells, HEK293 T cells, or BeWo cells. The cells can be used to produce heVLPs as described herein.

(14) In some embodiments, the methods include using cells that have or have not been manipulated to express any exogenous proteins except for a HERV envelope (e.g., as shown in Table 1), and, if desired, a plasma membrane recruitment domain (e.g., as shown in Table 6). In this embodiment, the “empty” particles that are produced can be loaded with biomolecule or chemical molecule cargo by utilizing nucleofection, lipid, polymer, or CaCl₂ transfection, sonication, freeze thaw, and/or heat shock of purified particles mixed with cargo. In all embodiments, producer cells do not express any human exogenous gag protein. This type of loading allows for cargo to be unmodified by fusions to plasma membrane recruitment domains and represents a significant advancement from previous VLP technology.

(15) In another embodiment, heVLPs that contain cargo are produced and isolated can be loaded with additional biomolecule or chemical molecule cargo by utilizing nucleofection, lipid, polymer, or CaCl₂ transfection, sonication, freeze thaw, incubation at various temperatures, and/or heat shock of purified particles mixed with cargo.

(16) In some embodiments, the cargo is a therapeutic or diagnostic protein or nucleic acid encoding a therapeutic or diagnostic protein.

(17) In some embodiments, the cargo is a chemical compound or molecule.

(18) In some embodiments, the chemical molecule is a trigger for protein-protein dimerization or multimerization, such as the A/C heterodimerizer or rapamycin.

(19) In some embodiments, the chemical compound is a DNA PK inhibitor, such as M3814, NU7026, or NU7441 which potently enhance homology directed repair gene editing.

- (20) In some embodiments, the biomolecule cargo is a gene editing reagent.
- (21) In some embodiments, the gene editing reagent comprises a zinc finger (ZF), transcription activator-like effector (TALE), and/or CRISPR-based genome editing or modulating protein; a nucleic acid encoding a zinc finger (ZF), transcription activator-like effector (TALE), and/or CRISPR-based genome editing or modulating protein; or a ribonucleoprotein complex (RNP) comprising a CRISPR-based genome editing or modulating protein.
- (22) In some embodiments, the gene editing reagent is selected from the proteins listed in Tables 2, 3, 4 & 5.
- (23) In some embodiments, the gene editing reagent comprises a CRISPR-based genome editing or modulating protein, and the heVLP further comprises one or more guide RNAs that bind to and direct the CRISPR-based genome editing or modulating protein to a target sequence.
- (24) In some embodiments, the cargo comprises a covalent or non-covalent connection to a human-endogenous GAG or other plasma membrane recruitment domain, preferably as shown in Table 6. Covalent connections, for example, can include direct protein-protein fusions generated from a single reading frame, inteins that can form peptide bonds, other proteins that can form covalent connections at R-groups and/or RNA splicing. Non-covalent connections, for example, can include DNA/DNA, DNA/RNA, and/or RNA/RNA hybrids (nucleic acids base pairing to other nucleic acids via hydrogen-bonding interactions), protein domains that dimerize or multimerize with or without the need for a chemical compound/molecule to induce the protein-protein binding, single chain variable fragments, nanobodies, affibodies, proteins that bind to DNA and/or RNA, proteins with quaternary structural interactions, optogenetic protein domains that can dimerize or multimerize in the presence of certain light wavelengths, and/or naturally reconstituting split proteins.
- (25) In some embodiments, the cargo comprises a fusion to a dimerization domain or protein-protein binding domain that may or may not require a molecule to trigger dimerization or protein-protein binding.
- (26) In some embodiments, the producer cells are FDA-approved cells lines, allogenic cells, and/or autologous cells derived from a donor.
- (27) In some embodiments, the full or active peptide domains of human CD47 may be incorporated in the heVLP surface to reduce immunogenicity.
- (28) Examples of AAV proteins included here are AAV REP 52, REP 78, and VP1-3. The capsid site where proteins can be inserted is T138 starting from the VP1 amino acid counting. Dimerization domains could be inserted at this point in the capsid, for instance.
- (29) Examples of dimerization domains included here that may or may not need a small molecule inducer are dDZF1, dDZF2, DmrA, DmrB, DmrC, FKBP, FRB, GCN4 scFv, 10×/24×GCN4, GFP nanobody and GFP.
- (30) Examples of split inteins included here are Npu DnaE, Cfa, Vma, and Ssp DnaE.
- (31) Examples of other split proteins included here that make a covalent bond together are Spy Tag and Spy Catcher.
- (32) Examples of RNA binding proteins included here are MS2, Com, and PP7.
- (33) Examples of synthetic DNA-binding zinc fingers included here are ZF6/10, ZF8/7, ZF9, MK10, Zinc Finger 268, and Zinc Finger 268/NRE.
- (34) Examples of proteins that multimerize as a result of quaternary structure included here are *E. coli* ferritin, and the other chimeric forms of ferritin.
- (35) Examples of optogenetic “light-inducible proteins” included here are Cry2, CIBN, and Lov2-Ja.
- (36) Examples of peptides the enhance transduction included here are L17E, Vectofusin, KALA, and the various forms of nisin.
- (37) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs.

Methods and materials are described herein for use in the present invention; other, suitable methods and materials known in the art can also be used. The materials, methods, and examples are illustrative only and not intended to be limiting. All publications, patent applications, patents, sequences, database entries, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. (38) Other features and advantages of the invention will be apparent from the following detailed description and figures, and from the claims.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1: Depiction of exemplary T2heVLP/T4heVLP production and transduction for RNP/protein delivery. All heVLP expression constructs are stably integrated in the genome of the producer cell. Construct 1-0 corresponds to the human-endogenous GAG (hGAG). Construct 1-1 corresponds to the human-endogenous GAG or other phospholipid bilayer recruitment domain. 1-2 corresponds to the cargo. 2 corresponds to an optional guide RNA. 1-0, 1-1 and 1-2 are translated in the cytosol where the fusion of 1-1 and 1-2 complexes with guide RNA before it is recruited to the phospholipid bilayer. 3 corresponds to a HERV-derived glycoprotein (hENV). The HERV-derived glycoprotein is expressed as a transmembrane protein on the plasma membrane. hGAG drives budding of cargo-containing heVLPs from the plasma membrane to extracellular space. These particles are purified and are able to fuse with target cells and deliver cargo by interacting with surface receptors at the target cell surface.
- (2) FIG. 2: Depiction of purified heVLPs entering a target cell and delivering cargo to the cytosol. Importantly, the human-endogenous GAG or other phospholipid bilayer recruitment domain allows cargo to enter the target cell nucleus as long as cargo possesses a nuclear localization sequence.
- (3) FIG. 3: Exemplary T1heVLP-delivered spCas9 genome editing in vitro. HEK 293T cells transduced with T1heVLPs containing PLC PH fused to spCas9, hGAGK.sub.con fused to spCas9, or human Activity-regulated cytoskeleton-associated protein (hArc) fused to spCas9 targeted to VEGF site #3. heVLPs are pseudotyped with either hENVW (left chart) or hENVFRD (right chart). Gene modification is measured by amplicon sequencing.
- (4) FIG. 4: Depiction of T1heVLP/T3heVLP production. Plasmid DNA constructs involved in the transfection encode cargo, an optional guide RNA, hGAG and a HERV-derived glycoprotein. Plasmids, or other types of DNA molecules, will be distributed throughout the production cell, so constructs located in the nucleus will express heVLP components and cargo, and constructs located near the plasma membrane or endosomes will be encapsulated within budding heVLPs.
- (5) FIG. 5: Depiction of exemplary heVLP and cargo configuration
- (6) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo or particles being loaded by various particle loading methods described herein, such as electroporation.
- (7) FIG. 6: Depiction of exemplary heVLP and cargo configuration
- (8) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo-gag fusion or particles being loaded by various particle loading methods described herein, such as electroporation.
- (9) FIG. 7: Depiction of exemplary heVLP and cargo configuration
- (10) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo-PH fusion or particles being loaded by various particle loading methods described herein, such as electroporation.
- (11) FIG. 8: Depiction of exemplary heVLP and cargo configuration
- (12) This particle was created by producer cells expressing an envelope protein. Cargo was

packaged inside the particle either by producer cells expressing cargo-gag/PH fusion or particles being loaded by various particle loading methods described herein, such as electroporation.

(13) FIG. 9: Depiction of exemplary heVLP and cargo configuration

(14) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of a dimerization molecule (A/C heterodimerizer) either by producer cells expressing cargo and gag fused to DmrA or DmrC or particles being loaded by various particle loading methods described herein, such as electroporation.

(15) FIG. 10: Depiction of exemplary heVLP and cargo configuration

(16) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of a dimerization molecule (A/C heterodimerizer) either by producer cells expressing cargo and PH fused to DmrA or DmrC or particles being loaded by various particle loading methods described herein, such as electroporation.

(17) FIG. 11: Depiction of exemplary heVLP and cargo configuration

(18) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of a dimerization molecule (A/C heterodimerizer) either by producer cells expressing cargo and gag/PH fused to DmrA or DmrC or particles being loaded by various particle loading methods described herein, such as electroporation.

(19) FIG. 12: Depiction of exemplary heVLP and cargo configuration

(20) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo and gag fused to an RNA binding protein (RBP), MS2, that binds to its MS2 RNA stem loop (MS2 SL) that is complexed with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(21) FIG. 13: Depiction of exemplary heVLP and cargo configuration

(22) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo and PH fused to an RNA binding protein (RBP), MS2, that binds to its RNA stem loop (MS2 SL) that is complexed with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(23) FIG. 14: Depiction of exemplary heVLP and cargo configuration

(24) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo and gag/PH fused to an RNA binding protein (RBP), MS2, that binds to its RNA stem loop (MS2 SL) that is complexed with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(25) FIG. 15: Depiction of exemplary heVLP and cargo configuration

(26) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of dimerization molecule (A/C Heterodimerizer) either by producer cells expressing cargo and gag and an RNA binding protein (RBP), MS2, fused to DmrA or DmrC that binds to its RNA stem loop (MS2 SL) that is complexed with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(27) FIG. 16: Depiction of exemplary heVLP and cargo configuration

(28) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of dimerization molecule (A/C Heterodimerizer) either by producer cells expressing cargo and PH and an RNA binding protein (RBP), MS2, fused to DmrA or DmrC that binds to its RNA stem loop (MS2 SL) that is complexed with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(29) FIG. 17: Depiction of exemplary heVLP and cargo configuration

(30) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of dimerization molecule (A/C Heterodimerizer) either by producer cells expressing cargo and gag/PH and an RNA binding protein (RBP), MS2, fused to DmrA or DmrC that binds to its RNA stem loop (MS2 SL) that is complexed with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(31) FIG. 18: Depiction of exemplary heVLP and cargo configuration

(32) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo and gag fused to a repetitive GCN4 domain that is bound by an scFv that is fused with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(33) FIG. 19: Depiction of exemplary heVLP and cargo configuration

(34) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo and PH fused to a repetitive GCN4 domain that is bound by an scFv that is fused with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(35) FIG. 20: Depiction of exemplary heVLP and cargo configuration

(36) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle either by producer cells expressing cargo and gag/PH fused to a repetitive GCN4 domain that is bound by an scFv that is fused with cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(37) FIG. 21: Depiction of exemplary heVLP and cargo configuration

(38) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of a dimerization molecule (A/C Heterodimerizer) by producer cells expressing gag and a repetitive GCN4 domain that are fused to DmrA or DmrC. GCN4 is bound by an scFv that is fused with cargo that is also being expressed in producer cells. Particles could also be loaded by various particle loading methods described herein, such as electroporation.

(39) FIG. 22: Depiction of exemplary heVLP and cargo configuration

(40) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of a dimerization molecule (A/C Heterodimerizer) by producer cells expressing PH and a repetitive GCN4 domain that are fused to DmrA or DmrC. GCN4 is bound by an scFv that is fused with cargo that is also being expressed in producer cells. Particles could also be loaded by various particle loading methods described herein, such as electroporation.

(41) FIG. 23: Depiction of exemplary heVLP and cargo configuration

(42) This particle was created by producer cells expressing an envelope protein. Cargo was packaged inside the particle in the presence of a dimerization molecule (A/C Heterodimerizer) by producer cells expressing gag/PH and a repetitive GCN4 domain that are fused to DmrA or DmrC. GCN4 is bound by an scFv that is fused with cargo that is also being expressed in producer cells. Particles could also be loaded by various particle loading methods described herein, such as electroporation.

(43) FIG. 24: Depiction of exemplary heVLP and cargo configuration

(44) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles) was packaged inside the particle either by producer cells expressing cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(45) FIG. 25: Depiction of exemplary heVLP and cargo configuration

(46) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles) was packaged inside the particle either by producer cells expressing cargo and gag or particles being loaded by various particle loading methods described herein, such as

electroporation.

(47) FIG. 26: Depiction of exemplary heVLP and cargo configuration

(48) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles) was packaged inside the particle either by producer cells expressing cargo and PH or particles being loaded by various particle loading methods described herein, such as electroporation.

(49) FIG. 27: Depiction of exemplary heVLP and cargo configuration

(50) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles) was packaged inside the particle either by producer cells expressing cargo and gag/PH or particles being loaded by various particle loading methods described herein, such as electroporation.

(51) FIG. 28: Depiction of exemplary heVLP and cargo configuration

(52) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles with DmrB inserted in the Capsid protein, VP2) was packaged inside the particle in the presence of DmrB dimerizer molecule either by producer cells expressing cargo and gag fused to DmrB or particles being loaded by various particle loading methods described herein, such as electroporation.

(53) FIG. 29: Depiction of exemplary heVLP and cargo configuration

(54) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles with DmrB inserted in the Capsid protein, VP2) was packaged inside the particle in the presence of DmrB dimerizer molecule either by producer cells expressing cargo and PH fused to DmrB or particles being loaded by various particle loading methods described herein, such as electroporation.

(55) FIG. 30: Depiction of exemplary heVLP and cargo configuration

(56) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles with DmrB inserted in the Capsid protein, VP2) was packaged inside the particle in the presence of DmrB dimerizer molecule either by producer cells expressing cargo and gag/PH fused to DmrB or particles being loaded by various particle loading methods described herein, such as electroporation.

(57) FIG. 31: Depiction of exemplary heVLP and cargo configuration

(58) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles with DmrB inserted in the Capsid protein, VP2) was packaged inside the particle in the presence of DmrB dimerizer and A/C Heterodimerizer molecules either by producer cells expressing cargo and gag fused to DmrA, DmrB, or DmrC, or particles being loaded by various particle loading methods described herein, such as electroporation.

(59) FIG. 32: Depiction of exemplary heVLP and cargo configuration

(60) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles with DmrB inserted in the Capsid protein, VP2) was packaged inside the particle in the presence of DmrB dimerizer and A/C Heterodimerizer molecules either by producer cells expressing cargo and PH fused to DmrA, DmrB, or DmrC, or particles being loaded by various particle loading methods described herein, such as electroporation.

(61) FIG. 33: Depiction of exemplary heVLP and cargo configuration

(62) This particle was created by producer cells expressing an envelope protein. Cargo (AAV particles with DmrB inserted in the Capsid protein, VP2) was packaged inside the particle in the presence of DmrB dimerizer and A/C Heterodimerizer molecules either by producer cells expressing cargo and gag/PH fused to DmrA, DmrB, or DmrC, or particles being loaded by various particle loading methods described herein, such as electroporation.

(63) FIG. 34: Depiction of exemplary heVLP and cargo configuration

(64) This particle was created by producer cells expressing an envelope protein. Cargo (single-stranded DNA) can be packaged inside the particle by various particle loading methods described

herein, such as electroporation.

(65) FIG. 35: Depiction of exemplary heVLP and cargo configuration

(66) This particle was created by producer cells expressing an envelope protein and gag. Cargo (single-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(67) FIG. 36: Depiction of exemplary heVLP and cargo configuration

(68) This particle was created by producer cells expressing an envelope protein and PH. Cargo (single-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(69) FIG. 37: Depiction of exemplary heVLP and cargo configuration

(70) This particle was created by producer cells expressing an envelope protein and gag/PH. Cargo (single-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(71) FIG. 38: Depiction of exemplary heVLP and cargo configuration

(72) This particle was created by producer cells expressing an envelope protein. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(73) FIG. 39: Depiction of exemplary heVLP and cargo configuration

(74) This particle was created by producer cells expressing an envelope protein and gag. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(75) FIG. 40: Depiction of exemplary heVLP and cargo configuration

(76) This particle was created by producer cells expressing an envelope protein and PH. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(77) FIG. 41: Depiction of exemplary heVLP and cargo configuration

(78) This particle was created by producer cells expressing an envelope protein and gag/PH. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(79) FIG. 42: Depiction of exemplary heVLP and cargo configuration

(80) This particle was created by producer cells expressing an envelope protein and gag fused to a zinc finger protein (ZFP) that will bind a specific sequence in the cargo. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(81) FIG. 43: Depiction of exemplary heVLP and cargo configuration

(82) This particle was created by producer cells expressing an envelope protein and PH fused to a zinc finger protein (ZFP) that will bind a specific sequence in the cargo. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(83) FIG. 44: Depiction of exemplary heVLP and cargo configuration

(84) This particle was created by producer cells expressing an envelope protein and gag/PH fused to a zinc finger protein (ZFP) that will bind a specific sequence in the cargo. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(85) FIG. 45: Depiction of exemplary heVLP and cargo configuration

(86) This particle was created by producer cells expressing an envelope protein and gag and a zinc finger protein (ZFP) that will bind a specific sequence in the cargo fused to DmrA or DmrC in the presence of A/C Heterodimerizer molecule. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(87) FIG. 46: Depiction of exemplary heVLP and cargo configuration

(88) This particle was created by producer cells expressing an envelope protein and PH and a zinc finger protein (ZFP) that will bind a specific sequence in the cargo fused to DmrA or DmrC in the presence of A/C Heterodimerizer molecule. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(89) FIG. 47: Depiction of exemplary heVLP and cargo configuration

(90) This particle was created by producer cells expressing an envelope protein and gag/PH and a zinc finger protein (ZFP) that will bind a specific sequence in the cargo fused to DmrA or DmrC in the presence of A/C Heterodimerizer molecule. Cargo (double-stranded DNA) can be packaged inside the particle by various particle loading methods described herein, such as electroporation.

(91) FIG. 48: Depiction of exemplary heVLP and cargo configuration

(92) This particle was created by producer cells expressing an envelope protein and gag fused to a zinc finger protein (ZFP) that will bind a specific sequence in the cargo. Cargo (double-stranded DNA bound by Cas9 RNP-ZFP fusion) can be packaged inside the particle by various particle loading methods described herein, such as electroporation. Alternatively, the Cas9 RNP-ZFP fusion could be expressed by the producer cells and the particles could be loaded by various particle loading methods described herein, such as electroporation.

(93) FIG. 49: Depiction of exemplary heVLP and cargo configuration

(94) This particle was created by producer cells expressing an envelope protein and PH fused to a zinc finger protein (ZFP) that will bind a specific sequence in the cargo. Cargo (double-stranded DNA bound by Cas9 RNP-ZFP fusion) can be packaged inside the particle by various particle loading methods described herein, such as electroporation. Alternatively, the Cas9 RNP-ZFP fusion could be expressed by the producer cells and the particles could be loaded by various particle loading methods described herein, such as electroporation.

(95) FIG. 50: Depiction of exemplary heVLP and cargo configuration

(96) This particle was created by producer cells expressing an envelope protein and gag/PH fused to a zinc finger protein (ZFP) that will bind a specific sequence in the cargo. Cargo (double-stranded DNA bound by Cas9 RNP-ZFP fusion) can be packaged inside the particle by various particle loading methods described herein, such as electroporation. Alternatively, the Cas9 RNP-ZFP fusion could be expressed by the producer cells and the particles could be loaded by various particle loading methods described herein, such as electroporation.

(97) FIG. 51: Depiction of exemplary heVLP and cargo configuration

(98) This particle was created by producer cells expressing an envelope protein and gag fused to a zinc finger protein (ZFP) fused to DmrA or DmrC that will bind a specific sequence in the cargo in the presence of A/C Heterodimerizer molecule. Cargo (double-stranded DNA bound by Cas9 RNP-ZFP fusion) can be packaged inside the particle by various particle loading methods described herein, such as electroporation. Alternatively, the Cas9 RNP-ZFP fusion could be expressed by the producer cells and the particles could be loaded by various particle loading methods described herein, such as electroporation.

(99) FIG. 52: Depiction of exemplary heVLP and cargo configuration

(100) This particle was created by producer cells expressing an envelope protein and PH fused to a zinc finger protein (ZFP) fused to DmrA or DmrC that will bind a specific sequence in the cargo in the presence of A/C Heterodimerizer molecule. Cargo (double-stranded DNA bound by Cas9 RNP-ZFP fusion) can be packaged inside the particle by various particle loading methods described herein, such as electroporation. Alternatively, the Cas9 RNP-ZFP fusion could be expressed by the producer cells and the particles could be loaded by various particle loading methods described herein, such as electroporation.

(101) FIG. 53: Depiction of exemplary heVLP and cargo configuration

(102) This particle was created by producer cells expressing an envelope protein and gag/PH fused to a zinc finger protein (ZFP) fused to DmrA or DmrC that will bind a specific sequence in the cargo in the presence of A/C Heterodimerizer molecule. Cargo (double-stranded DNA bound by

Cas9 RNP-ZFP fusion) can be packaged inside the particle by various particle loading methods described herein, such as electroporation. Alternatively, the Cas9 RNP-ZFP fusion could be expressed by the producer cells and the particles could be loaded by various particle loading methods described herein, such as electroporation.

(103) FIG. 54: Depiction of exemplary heVLP and cargo configuration

(104) This particle was created by producer cells expressing an envelope protein. Cargo (RNA) was packaged inside the particle either by producer cells expressing cargo or particles being loaded by various particle loading methods described herein, such as electroporation.

(105) FIG. 55: Depiction of exemplary heVLP and cargo configuration

(106) This particle was created by producer cells expressing an envelope protein. Cargo (RNA) was packaged inside the particle either by producer cells expressing cargo and gag or particles being loaded by various particle loading methods described herein, such as electroporation.

(107) FIG. 56: Depiction of exemplary heVLP and cargo configuration

(108) This particle was created by producer cells expressing an envelope protein. Cargo (RNA) was packaged inside the particle either by producer cells expressing cargo and PH or particles being loaded by various particle loading methods described herein, such as electroporation.

(109) FIG. 57: Depiction of exemplary heVLP and cargo configuration

(110) This particle was created by producer cells expressing an envelope protein. Cargo (RNA) was packaged inside the particle either by producer cells expressing cargo and gag/PH or particles being loaded by various particle loading methods described herein, such as electroporation.

(111) FIG. 58: Depiction of exemplary heVLP and cargo configuration

(112) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with MS2 stem loop(s)) was packaged inside the particle either by producer cells expressing cargo and gag fused to MS2 or particles being loaded by various particle loading methods described herein, such as electroporation.

(113) FIG. 59: Depiction of exemplary heVLP and cargo configuration

(114) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with MS2 stem loop(s)) was packaged inside the particle either by producer cells expressing cargo and PH fused to MS2 or particles being loaded by various particle loading methods described herein, such as electroporation.

(115) FIG. 60: Depiction of exemplary heVLP and cargo configuration

(116) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with MS2 stem loop(s)) was packaged inside the particle either by producer cells expressing cargo and gag/PH fused to MS2 or particles being loaded by various particle loading methods described herein, such as electroporation.

(117) FIG. 61: Depiction of exemplary heVLP and cargo configuration

(118) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with MS2 stem loop(s)) was packaged inside the particle either by producer cells expressing cargo and gag and MS2 fused to DmrA or DmrC in the presence of A/C heterodimerizer, or particles being loaded by various particle loading methods described herein, such as electroporation.

(119) FIG. 62: Depiction of exemplary heVLP and cargo configuration

(120) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with MS2 stem loop(s)) was packaged inside the particle either by producer cells expressing cargo and PH and MS2 fused to DmrA or DmrC in the presence of A/C heterodimerizer, or particles being loaded by various particle loading methods described herein, such as electroporation.

(121) FIG. 63: Depiction of exemplary heVLP and cargo configuration

(122) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with MS2 stem loop(s)) was packaged inside the particle either by producer cells expressing cargo and gag/PH and MS2 fused to DmrA or DmrC in the presence of A/C heterodimerizer, or particles being loaded by various particle loading methods described herein, such as electroporation.

(123) FIG. **64**: Depiction of exemplary heVLP and cargo configuration

(124) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with RBP stem loop(s)) was packaged inside the particle either by producer cells expressing cargo fused to an RBP and gag fused to another RBP or particles being loaded by various particle loading methods described herein, such as electroporation.

(125) FIG. **65**: Depiction of exemplary heVLP and cargo configuration

(126) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with RBP stem loop(s)) was packaged inside the particle either by producer cells expressing cargo fused to an RBP and PH fused to another RBP or particles being loaded by various particle loading methods described herein, such as electroporation.

(127) FIG. **66**: Depiction of exemplary heVLP and cargo configuration

(128) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with RBP stem loop(s)) was packaged inside the particle either by producer cells expressing cargo fused to an RBP and gag/PH fused to another RBP or particles being loaded by various particle loading methods described herein, such as electroporation.

(129) FIG. **67**: Depiction of exemplary heVLP and cargo configuration

(130) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with RBP stem loop(s)) was packaged inside the particle either by producer cells expressing cargo fused to an RBP and gag and another RBP fused to DmrA or DmrC in the presence of A/C Heterodimerizer molecule, or particles being loaded by various particle loading methods described herein, such as electroporation.

(131) FIG. **68**: Depiction of exemplary heVLP and cargo configuration

(132) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with RBP stem loop(s)) was packaged inside the particle either by producer cells expressing cargo fused to an RBP and PH and another RBP fused to DmrA or DmrC in the presence of A/C Heterodimerizer molecule, or particles being loaded by various particle loading methods described herein, such as electroporation.

(133) FIG. **69**: Depiction of exemplary heVLP and cargo configuration

(134) This particle was created by producer cells expressing an envelope protein. Cargo (RNA with RBP stem loop(s)) was packaged inside the particle either by producer cells expressing cargo fused to an RBP and gag/PH and another RBP fused to DmrA or DmrC in the presence of A/C Heterodimerizer molecule, or particles being loaded by various particle loading methods described herein, such as electroporation.

DETAILED DESCRIPTION

(135) Therapeutic proteins and nucleic acids hold great promise, but for many of these large biomolecules delivery into cells is a hurdle to clinical development. Genome editing reagents such as zinc finger nucleases (ZFNs) or RNA-guided, enzymatically active/inactive DNA binding proteins such as Cas9 have undergone rapid advancements in terms of specificity and the types of edits that can be executed, but the hurdle of safe in vivo delivery still precludes efficacious gene editing therapies. The following details the characteristics of the heVLP that make it a novel and optimal platform for the delivery of genome editing reagents, and contrasts heVLPs with canonical delivery modalities.

(136) Retroviral particles, such as lentivirus, have been developed to deliver RNA that is reverse transcribed to DNA that may or may not be integrated into genomic DNA. VLPs have been developed that mimic virus particles in their ability to self-assemble, but are not infectious as they lack some of the core viral genes. Both lentiviral and VLP vectors are typically produced by transiently transfecting a producer cell line with plasmids that encode all components necessary to produce lentiviral particles or VLP. One major flaw that we have discovered regarding lentiviral particles and VSVG-based VLPs that are produced by this conventional transient transfection method is that, in addition to their conventional cargo, these particles package and deliver plasmid

DNA that was used in the initial transient transfection. This unintended plasmid DNA delivery can be immunogenic and cause undesirable effects, such as plasmid DNA being integrated into genomic DNA. It is important to specify the type of biomolecules and/or chemicals that are to be delivered within particles, and heVLPs have been designed to possess this germane capability.

(137) The heVLPs described herein can deliver DNA only, DNA+RNA+protein, or RNA+protein. Importantly, heVLPs are the first VLP delivery modality that leverages select components from human endogenous retroviruses (HERVs) to create particles for customizable cargo delivery into eukaryotic cells. heVLPs are capable of controlling the form of the cargo (DNA, protein, and/or RNA). All other previously described VLPs and viral particles package and deliver unwanted plasmid DNA (or other types of DNA-based gene expression constructs) introduced into particle producer cells via transient transfection in addition to the intended protein and/or RNA cargo(s).

(138) Another non-obvious aspect of heVLPs is the ENV protein on the surface of the heVLP. The ENV protein is responsible for the ability of heVLPs to efficiently deliver cargo into cells. The majority of retroviral ENV proteins require post-translational modifications in the form of proteolytic cleavage of the intracellular domain (ICD) of the ENV protein in order to activate the fusogenicity of the ENV protein; this is essential for infectivity.^{sup.1} The envelope proteins described in Table 1 are all derived from HERVs that are expressed to varying levels in healthy human tissues (or HERV ENV consensus sequences). Some of these sequences possess ICD truncations that have been shown to enhance fusogenicity, but most do not require truncation.

(139) heVLPs do not require exogenous, virally-derived GAG for particle formation because heVLPs utilize human-endogenous GAG proteins from HERVs (or HERV GAG consensus sequences).^{sup.1} These HERV GAG proteins enable heVLP formation and are expressed to varying levels in healthy human tissues. Importantly, heVLPs are different from previously described viral particles, VLPs, and extracellular vesicles because heVLPs are composed of a novel combination of HERV ENV and GAG components, and heVLPs lack components from exogenous viruses.^{sup.2,3} Because of the above mentioned design optimizations, heVLPs are particularly suited for delivery of DNA, RNA, protein, or combinations of biomolecules and/or chemicals, such as DNA-encoded or RNP-based genome editing reagents.

(140) Genome editing reagents, especially CRISPR-CAS, zinc finger, and TAL-nuclease-based reagents have the potential to become in vivo therapeutics for the treatment of genetic diseases, but techniques for delivering genome editing reagents into cells are severely limiting or unsafe for patients. Conventional therapeutic monoclonal antibody delivery is successful at utilizing direct injection for proteins. Unfortunately, strategies for direct injection of gene editing proteins, such as Cas9, are hampered by immunogenicity, degradation, ineffective cell specificity, and inability to cross the plasma membrane or escape endosomes/lysosomes.^{sup.4-10} More broad applications of protein therapy and gene editing could be achieved by delivering therapeutic protein cargo to the inside of cells. Cas9, for example, cannot efficiently cross the phospholipid bilayer to enter into cells, and has been shown to have innate and adaptive immunogenic potential.^{sup.4-8} Therefore, it is not practical or favorable to deliver Cas9 by direct injection or as an external/internal conjugate to lipid, protein or metal-based nanoparticles that have cytotoxic and immunogenic properties and often yield low levels of desired gene modifications.^{sup.9-20}

(141) Nanoparticles that encapsulate cargo are another delivery strategy that can be used to deliver DNA, protein, RNA and RNPs into cells.^{sup.9-18} Nanoparticles can be engineered for cell specificity and can trigger endocytosis and subsequent endosome lysis. However, nanoparticles can have varying levels of immunogenicity due to an artificially-derived vehicle shell.^{sup.9-20} Many nanoparticles rely on strong opposing charge distributions to maintain particle structural integrity, and the electrostatics can make it toxic and unfit for many in vivo therapeutic scenarios.^{sup.9} Nanoparticles that deliver RNA have had successes in recent clinical trials, but most have only been used to deliver siRNA or shRNA. Toxicity from such nanoparticles is still a major concern.^{sup.9} Nanoparticles that deliver mRNA coding for genome editing RNPs have also been a

recent success, but these create a higher number of off-target effects compared to protein delivery and RNA stability is lower than that of protein..^{sup.17} Nanoparticles that deliver genome editing RNPs and DNA have been a significant breakthrough because they can leverage both homology directed repair (HDR) and non-homologous end joining (NHEJ), but exhibit prohibitively low gene modification frequencies in vitro and in vivo, and therefore currently have limited applications in vivo as a gene editing therapeutic..^{sup.15}

(142) Currently, the clinical standard vehicles for delivering genome editing therapeutics are adeno-associated virus (AAV). Although AAV vectors are a promising delivery modality that can successfully deliver DNA into eukaryotic cells, AAV cannot efficiently package and deliver DNA constructs larger than 4.5 kb and this precludes delivery of many CRISPR-based gene editing reagents that require larger DNA expression constructs. CRISPR-based gene editing reagents can be split into multiple different AAV particles, but this strategy drastically reduces delivery and editing efficiency. Depending on the dose required, AAV and adenoviral vectors can have varying levels of immunogenicity. In addition, inverted-terminal repeats (ITRs) in the AAV DNA construct can promote the formation of spontaneous episomes leading to prolonged expression of genome editing reagents and increased off-target effects. ITRs can also promote the undesired integration of AAV DNA into genomic DNA..^{sup.21-24} Recently, VLPs have been utilized to deliver mRNA and protein cargo into the cytosol of cells..^{sup.2,3,25-30} VLPs have emerged as a substitute delivery modality for retroviral particles. VLPs can be designed to lack the ability to integrate retroviral DNA, and to package and deliver protein/RNP/DNA. However, most VLPs, including recently conceived VLPs that deliver genome editing reagents known to date, utilize HIV or other virally-derived gag-pol protein fusions and viral proteases to generate retroviral-like particles..^{sup.25-27,29,30} Secondly, some VLPs containing RGNs also must package and express guide RNAs from a lentiviral DNA transcript..^{sup.27} Thirdly, some VLPs require a viral protease in order to form functional particles and release genome editing cargo..^{sup.25-27,29} Since this viral protease recognizes and cleaves at multiple amino acid motifs, it can cause damage to the protein cargo which could be hazardous for therapeutic applications. Fourthly, most published VLP modalities that deliver genome editing proteins to date exhibit low in vitro and in vivo gene modification efficiencies due to low packaging and transduction efficiency..^{sup.25-27} Fifthly, the complex viral genomes utilized for these VLP components possess multiple reading frames and employ RNA splicing that could result in spurious fusion protein products being delivered..^{sup.25-27,29,30} Sixthly, the presence of reverse transcriptase, integrase, capsid and a virally-derived envelope protein in these VLPs is not ideal for most therapeutic applications because of immunogenicity and off target editing concerns. Lastly, most retroviral particles, such as lentiviral particles, are pseudotyped with VSVG and nearly all described VLPs that deliver genome editing reagents hitherto possess and rely upon VSVG..^{sup.2,3,25-30} We have discovered that VSVG-based particles that are formed by transiently transfecting producer cells package and deliver DNA that was transfected. The current versions of VSVG-based VLPs cannot prevent this inadvertent delivery of DNA and this impedes the use of VLPs in scenarios that necessitate minimal immunogenicity and off target effects.

(143) Extracellular vesicles are another delivery modality that can package and deliver cargo within exosomes and ectosomes..^{sup.31,32} Similar to VLPs, extracellular vesicles are comprised of a phospholipid bilayer from a mammalian cell. Unlike VLPs, extracellular vesicles lack viral components and therefore have limited immunogenicity. Whereas VLPs have a great ability to enter cells due to external fusogenic glycoproteins (VSVG) extracellular vesicles mainly rely on cellular uptake via micropinocytosis and this limits the delivery efficiency of extracellular vesicles.

(144) heVLPs try to leverage the delivery benefits of extracellular vesicles and VLPs. heVLPs are the first VLP modality to eliminate all the potentially harmful exogenous, virally-derived components. heVLP components are known to be involved in extracellular vesicle biogenesis, they are known to possess local immunosuppressive properties, and their expression in healthy human

tissues minimizes the chance of eliciting an immune response because of central tolerance..^{sup.1} heVLPs are a safer and more effective alternative than previously described VLPs, extracellular vesicles, AAVs and nanoparticles-especially for delivery of genome editing reagents-because heVLPs are comprised of all human-derived components, heVLPs have the ability to deliver DNA+RNP, or RNP alone while other previously described VLPs cannot prevent transient transfection DNA from being unintentionally packaged and delivered, heVLPs can deliver specialty DNA molecules while previously described VLPs, nanoparticles and AAVs cannot or do not, and heVLPs can be produced with cells that have been derived from patients (autologous heVLPs) and other FDA-approved cell lines (allogenic heVLPs) to further reduce the risks of adverse immune reactions. Here, we describe methods and compositions for producing, purifying, and administering heVLPs for in vitro and in vivo applications of genome editing, epigenome modulation, transcriptome editing and proteome modulation. The desired editing outcome depends on the therapeutic context and will require different gene editing reagents. *Streptococcus pyogenes* Cas9 (spCas9) and *acidaminococcus sp.* Cas12a (functionalize) are two of the most popular RNA-guided enzymes for editing that leverages NHEJ for introducing stop codons or deletions, or HDR for causing insertions..^{sup.34-36} Cas9-deaminase fusions, also known as base editors, are the current standard for precise editing of a single nucleotide without double stranded DNA cleavage..^{sup.37,38} Importantly, this invention provides a novel way of packaging and delivering reagents for applications of genome editing, epigenome modulation, transcriptome editing and proteome modulation. Importantly, this invention is also the first to address the phenomenon of inadvertent DNA delivery in VLPs and the first to control for the type of biomolecule to be delivered (DNA, RNA, and/or protein) thereby increasing the types of therapeutic in vivo genome modifications that are possible and minimizing deleterious off target effects.

(145) Section 1: heVLP-Mediated Delivery of Cargo Including DNAs, Proteins, Compounds, and RNAs

(146) Conventional VLPs that have been engineered to encapsulate and deliver protein-based cargo commonly fuse cargo to the INT or GAG polyprotein..^{sup.25-27,29,30,39,40} After transient transfection of production plasmid DNA constructs, these protein fusions are translated in the cytosol of conventional VLP production cell lines, the gag matrix is acetylated and recruited to the cell membrane, and the gag fusions are encapsulated (transient transfection DNA is also unintentionally encapsulated) within VLPs as VLPs bud off of the membrane into extracellular space.

(147) In contrast, the heVLPs described herein can package protein-based cargo by integrating all production DNA into the genomic DNA of production cell lines. Once cell lines are created, protein delivery heVLPs can be produced in a constitutive or inducible fashion. Proteins are packaged into heVLP by fusing select human-endogenous GAG proteins or other plasma membrane recruitment domains to protein-based cargo (e.g., as shown in Table 6). Human-endogenous GAG proteins and human pleckstrin homology (PH) domains localize to biological membranes. PH domains interact with phosphatidylinositol lipids and proteins within biological membranes, such as PIP2, PIP3, $\beta\gamma$ -subunits of GPCRs, and PKC..^{sup.41,42} However, in addition to localizing to phospholipid bilayers, human-endogenous GAG proteins drive budding and particle formation..^{sup.42} This dual functionality of human-endogenous GAG enables packaging of cargo and budding/formation of particles. One such human-endogenous GAG protein used for this purpose is the human Arc protein, which can be fused to protein-based cargo to recruit cargo to the cytosolic side of the phospholipid bilayer..^{sup.43} These human-endogenous GAG phospholipid bilayer recruitment domains can be fused to the N-terminus or C-terminus of protein-based cargo via polypeptide linkers of variable length regardless of the location or locations of one or more nuclear localization sequence(s) (NLS) within the cargo. Preferably, the linker between protein-based cargo and the human-endogenous GAG phospholipid bilayer recruitment domain is a polypeptide linker 5-20, e.g., 8-12, e.g., 10, amino acids in length primarily composed of glycines and serines. The human-

endogenous GAG or other phospholipid bilayer recruitment domain localizes the cargo to the phospholipid bilayer and this protein cargo is packaged within heVLPs that bud off from the producer cell into extracellular space (FIG. 1). In this application, the use of these human-endogenous GAG and other phospholipid bilayer recruitment domains is novel and unique in that these human-endogenous GAG and other proteins can facilitate for localization of cargo to the cytosolic face of the plasma membrane within the heVLP production cells, and they also allow for cargo to localize to the nucleus of heVLP-transduced cells without the utilization of exogenous retroviral GAG or chemical and/or light-based dimerization systems (FIG. 2). The heVLP delivery of Cas9, for example, is significantly more efficient with a fusion to a human-endogenous GAG protein compared to a fusion to a PH plasma membrane recruitment domain or no fusion at all (FIG. 3).

(148) heVLPs can also package and deliver a combination of DNA and RNA if heVLPs are produced via transient transfection of a production cell line. DNA that is transfected into cells will possess size-dependent mobility such that a fraction of the transfected DNA will remain in the cytosol while another fraction of the transfected DNA will localize to the nucleus..sup.44-46 One fraction of the transfected DNA in the nucleus will expressed components needed to create heVLPs and the other fraction in the cytosol/near the plasma membrane will be encapsulated and delivered in heVLPs (FIG. 4).

(149) heVLP “Cargo” as used herein can refer to a one or more of chemicals, e.g., small molecule compounds, combination of DNA, RNA, and protein, a combination of RNA and protein, a combination of DNA and protein, or protein, e.g., for therapeutic or diagnostic use, or for the applications of genome editing, epigenome modulation, and/or transcriptome modulation. In addition, endogenous RNA and protein from the producer cells get packaged and/or incorporated into heVLPs. In order to simplify these distinctions, a combination of exogenous DNA, exogenous RNA, and protein (exogenous and/or endogenous protein) will be referred to as type 1 cargo (T1heVLPs), exogenous RNA and protein (exogenous and/or endogenous protein) will be referred to as type 2 cargo (T2heVLPs), a combination of exogenous DNA and proteins (exogenous and/or endogenous protein) will be referred to as type 3 cargo (T3heVLPs), proteins (exogenous and/or endogenous protein) will be referred to as type 4 cargo (T4heVLPs). Therefore, T1 contains DNA, RNA, +/-exogenous protein, T2 contains RNA+/-exogenous protein, T3 contains DNA+/-exogenous protein, and T4 is a particle with or without exogenous protein cargo. Hence, T4 without exogenous protein is considered an “empty particle” because there is no “exogenous cargo.” “Exogenous cargo” is cargo not endogenous to the producer cells that can be packaged and/or incorporated into heVLPs. In addition, T1-T4heVLPs can package exogenous chemical molecules in addition to the types of cargoes present in T1-T4heVLPs. RNA in this context, for example, could be single guide RNA (sgRNA), Clustered Regularly Interspaced Palindromic Repeat (CRISPR) RNA (crRNA), and/or mRNA coding for cargo.

(150) As used herein, “small molecules” refers to small organic or inorganic molecules of molecular weight below about 3,000 Daltons. In general, small molecules useful for the invention have a molecular weight of less than 3,000 Daltons (Da). The small molecules can be, e.g., from at least about 100 Da to about 3,000 Da (e.g., between about 100 to about 3,000 Da, about 100 to about 2500 Da, about 100 to about 2,000 Da, about 100 to about 1,750 Da, about 100 to about 1,500 Da, about 100 to about 1,250 Da, about 100 to about 1,000 Da, about 100 to about 750 Da, about 100 to about 500 Da, about 200 to about 1500, about 500 to about 1000, about 300 to about 1000 Da, or about 100 to about 250 Da).

(151) The cargo is limited by the diameter of the particles, e.g., which in some embodiments range from 150 nm to 500 nm. Cargo developed for applications of genome editing also includes nucleases and base editors. Nucleases include FokI and AcuI ZFNs and Transcription activator-like effector nucleases (TALENs) and CRISPR based nucleases or a functional derivative thereof (e.g., as shown in Table 2) (ZFNs are described, for example, in United States Patent Publications

20030232410; 20050208489; 20050026157; 20050064474; 20060188987; 20060063231; and International Publication WO 07/014275) (TALENs are described, for example, in United States Patent Publication U.S. Pat. No. 9,393,257B2; and International Publication WO2014134412A1) (CRISPR based nucleases are described, for example, in United States Patent Publications U.S. Pat. No. 8,697,359B1; US20180208976A1; and International Publications WO2014093661A2; WO2017184786A8)..sup.34-36 Base editors that are described by this work include any CRISPR based nuclease orthologs (wt, nickase, or catalytically inactive (CI)), e.g., as shown in Table 2, fused at the N-terminus to a deaminase or a functional derivative thereof (e.g., as shown in Table 3) with or without a fusion at the C-terminus to one or multiple uracil glycosylase inhibitors (UGIs) using polypeptide linkers of variable length (Base editors are described, for example, in United States Patent Publications US20150166982A1; US20180312825A1; U.S. Ser. No. 10/113,163B2; and International Publications WO2015089406A1; WO2018218188A2; WO2017070632A2; WO2018027078A8; WO2018165629A1)..sup.37,38 In addition, prime editors are also compatible with heVLP delivery modalities (Prime editors are described, for example, in Anzalone et al., Nature. 2019 December; 576(7785):149-157).

(152) sgRNAs complex with genome editing reagents during the packaging process and are co-delivered within heVLPs. To date, this concept has been validated in vitro by experiments that demonstrate the T2heVLP delivery of RGN RNP for the purposes of site specific editing of an endogenous site (FIG. 3). For example, T2heVLPs have been used to deliver Cas9 RNP to HEK 293T cells for the purposes of editing endogenous VEGF site #3 (FIG. 3).

(153) Cargo designed for the purposes of epigenome modulation includes the CI CRISPR based nucleases, zinc fingers (ZFs) and TALEs fused to an epigenome modulator or combination of epigenome modulators or a functional derivative thereof connected together by one or more variable length polypeptide linkers (Tables 2 & 4). T1-T4 cargo designed for the purposes of transcriptome editing includes CRISPR based nucleases or any functional derivatives thereof in Table 5 or CI CRISPR based nucleases or any functional derivatives thereof in Table 5 fused to deaminases in Table 3 by one or more variable length polypeptide linkers.

(154) The cargo can also include any therapeutically or diagnostically useful protein, DNA, RNP, or combination of DNA, protein and/or RNP. See, e.g., WO2014005219; U.S. Ser. No. 10/137,206; US20180339166; U.S. Pat. No. 5,892,020A; EP2134841B1; WO2007020965A1. For example, cargo encoding or composed of nuclease or base editor proteins or RNPs or derivatives thereof can be delivered to retinal cells for the purposes of correcting a splice site defect responsible for Leber Congenital Amaurosis type 10. In the mammalian inner ear, heVLP delivery of base editing reagents or HDR promoting cargo to sensory cells such as cochlear supporting cells and hair cells for the purposes of editing β -catenin (β -catenin Ser 33 edited to Tyr, Pro, or Cys) in order to better stabilize β -catenin could help reverse hearing loss.

(155) In another application, heVLP delivery of RNA editing reagents or proteome perturbing reagents could cause a transitory reduction in cellular levels of one or more specific proteins of interest (potentially at a systemic level, in a specific organ or a specific subset of cells, such as a tumor), and this could create a therapeutically actionable window when secondary drug(s) could be administered (this secondary drug is more effective in the absence of the protein of interest or in the presence of lower levels of the protein of interest). For example, heVLP delivery of RNA editing reagents or proteome perturbing reagents could trigger targeted degradation of MAPK and PI3K/AKT proteins and related mRNAs in vemurafenib/dabrafenib-resistant BRAF-driven tumor cells, and this could open a window for the administration of vemurafenib/dabrafenib because BRAF inhibitor resistance is temporarily abolished (resistance mechanisms based in the MAPK/PI3K/AKT pathways are temporarily downregulated by heVLP cargo). This example is especially pertinent when combined with heVLPs that are antigen inducible and therefore specific for tumor cells.

(156) In another application, heVLPs could deliver Yamanaka factors Oct3/4, Sox2, Klf4, and c-

Myc to human or mouse fibroblasts in order to generate induced pluripotent stem cells.

(157) In another application, heVLPs could deliver dominant-negative forms of proteins in order to elicit a therapeutic effect.

(158) heVLPs that are antigen-specific could be targeted to cancer cells in order to deliver proapoptotic proteins BIM, BID, PUMA, NOXA, BAD, BIK, BAX, BAK and/or HRK in order to trigger apoptosis of cancer cells.

(159) 90% of pancreatic cancer patients present with unresectable disease. Around 30% of patients with unresectable pancreatic tumors will die from local disease progression, so it is desirable to treat locally advanced pancreatic tumors with ablative radiation, but the intestinal tract cannot tolerate high doses of radiation needed to cause tumor ablation. Selective radioprotection of the intestinal tract enables ablative radiation therapy of pancreatic tumors while minimizing damage done to the surrounding gastrointestinal tract. To this end, heVLPs could be loaded with dCas9 fused to the transcriptional repressor KRAB and guide RNA targeting EGLN. EGLN inhibition has been shown to significantly reduce gastrointestinal toxicity from ablative radiation treatments because it causes selective radioprotection of the gastrointestinal tract but not the pancreatic tumor..sup.47

(160) Unbound steroid receptors reside in the cytosol. After binding to ligands, these receptors will translocate to the nucleus and initiate transcription of response genes. heVLPs could deliver single chain variable fragment (scFv) antibodies to the cytosol of cells that bind to and disrupt cytosolic steroid receptors. For example, the scFv could bind to the glucocorticoid receptor and prevent it from binding dexamethasone, and this would prevent transcription of response genes, such as metallothionein 1E which has been linked to tumorigenesis..sup.48

(161) heVLPs can be indicated for treatments that involve targeted disruption of proteins. For example, heVLPs can be utilized for targeting and disrupting proteins in the cytosol of cells by delivering antibodies/scFvs to the cytosol of cells. Classically, delivery of antibodies through the plasma membrane to the cytosol of cells has been notoriously difficult and inefficient. This mode of protein inhibition is similar to how a targeted small molecule binds to and disrupts proteins in the cytosol and could be useful for the treatment of a diverse array of diseases..sup.49-51

(162) In addition, the targeting of targeted small molecules is limited to proteins of a certain size that contain binding pockets which are relevant to catalytic function or protein-protein interactions. scFvs are not hampered by these limitations because scFvs can be generated that bind to many different moieties of a protein in order to disrupt catalysis and interactions with other proteins. For example, RAS oncoproteins are implicated across a multitude of cancer subtypes, and RAS is one of the most frequently observed oncogenes in cancer. For instance, the International Cancer Genome Consortium found KRAS to be mutated in 95% of their Pancreatic Adenocarcinoma samples. RAS isoforms are known to activate a variety of pathways that are dysregulated in human cancers, like the PI3K and MAPK pathways. Despite the aberrant roles RAS plays in cancer, no efficacious pharmacologic direct or indirect small molecule inhibitors of RAS have been developed and approved for clinical use. One strategy for targeting RAS could be heVLPs that can deliver specifically to cancer cells scFvs that bind to and disrupt the function of multiple RAS isoforms..sup.49-51

(163) FIGS. 5-69 provide exemplary heVLP configurations and non-limiting examples of cargo molecules.

(164) Section 2: heVLP Composition, Production, Purification and Applications

(165) heVLPs are produced from producer cell lines that are either transiently transfected with at least one plasmid or stably expressing constructs that have been integrated into the producer cell line genomic DNA. In some embodiments, for T1 and T3heVLPs, if a single plasmid is used in the transfection, it should comprise sequences encoding one or more HERV-derived glycoproteins (e.g., as shown in Table 1), one or more HERV-derived GAG proteins, cargo (e.g., a therapeutic protein or a gene editing reagent such as a zinc finger, transcription activator-like effector (TALE),

and/or CRISPR-based genome editing/modulating protein and/or RNP such as those found in Tables 2, 3, 4 & 5) with a fusion to a human-endogenous GAG or other plasma membrane recruitment domain (e.g., as shown in Table 6), and a guide RNA, if necessary. Preferably, two to three plasmids are used in the transfection. These two to three plasmids can include the following (any two or more can be combined in a single plasmid): 1. A plasmid comprising sequences encoding a therapeutic protein or a genome editing reagent, with a fusion to a human-endogenous GAG or other plasma membrane recruitment domain. 2. A plasmid comprising one or more HERV-derived glycoproteins (e.g., as listed in Table 1). 3. A plasmid comprising one or more HERV-derived GAG proteins. 4. If the genome editing reagent from plasmid 1 requires one or more guide RNAs, a plasmid comprising one or more guide RNAs apposite for the genome editing reagent in plasmid 1.

If it is desired to deliver a type of DNA molecule other than plasmid(s), the above-mentioned transfection can be performed with double-stranded closed-end linear DNA, episome, mini circle, double-stranded oligonucleotide and/or other specialty DNA molecules. Alternatively, for T2 and T4heVLPs, the producer cell line can be made to stably express the constructs (1 through 3) described in the transfection above.

(166) The plasmids, or other types of specialty DNA molecules described above, will also preferably include other elements to drive expression or translation of the encoded sequences, e.g., a promoter sequence; an enhancer sequence, e.g., 5' untranslated region (UTR) or a 3' UTR; a polyadenylation site; an insulator sequence; or another sequence that increases or controls expression (e.g., an inducible promoter element).

(167) Preferably, appropriate producer cell lines are primary or stable human cell lines refractory to the effects of transfection reagents and fusogenic effects due glycoproteins. Examples of appropriate cell lines include Human Embryonic Kidney (HEK) 293 cells, HEK293 T/17 SF cells kidney-derived Phoenix-AMPHO cells, and placenta-derived BeWo cells. For example, such cells could be selected for their ability to grow as adherent cells, or suspension cells. In some embodiments, the producer cells can be cultured in classical DMEM under serum conditions, serum-free conditions, or exosome-free serum conditions. T1 and T3heVLPs can be produced from cells that have been derived from patients (autologous heVLPs) and other FDA-approved cell lines (allogenic heVLPs) as long as these cells can be transfected with DNA constructs that encode the aforementioned heVLP production components by various techniques known in the art.

(168) In addition, if it is desirable, more than one genome editing reagent can be included in the transfection. The DNA constructs can be designed to overexpress proteins in the producer cell lines. The plasmid backbones, for example, used in the transfection can be familiar to those skilled in the art, such as the pCDNA3 backbone that employs the CMV promoter for RNA polymerase II transcripts or the U6 promoter for RNA polymerase III transcripts. Various techniques known in the art may be employed for introducing nucleic acid molecules into producer cells. Such techniques include chemical-facilitated transfection using compounds such as calcium phosphate, cationic lipids, cationic polymers, liposome-mediated transfection, such as cationic liposome like LIPOFECTAMINE (LIPOFECTAMINE 2000 or 3000 and TransIT-X2), polyethyleneimine, non-chemical methods such as electroporation, particle bombardment, or microinjection.

(169) A human producer cell line that stably expresses the necessary heVLP components in a constitutive and/or inducible fashion can be used for production of T2 and T4heVLPs. T2 and T4heVLPs can be produced from cells that have been derived from patients (autologous heVLPs) and other FDA-approved cell lines (allogenic heVLPs) if these cells have been converted into stable cell lines that express the aforementioned heVLP components.

(170) Also provided herein are the producer cells themselves.

(171) In some embodiments, in order for efficient recruitment of cargo into heVLPs, the cargo comprises a covalent or non-covalent connection to a human-endogenous GAG or other plasma membrane recruitment domain, preferably as shown in Table 6. Covalent connections, for example,

can include direct protein-protein fusions generated from a single reading frame, inteins that can form peptide bonds, other proteins that can form covalent connections at R-groups and/or RNA splicing..⁵²⁻⁵⁴ Non-covalent connections, for example, can include DNA/DNA, DNA/RNA, and/or RNA/RNA hybrids (nucleic acids base pairing to other nucleic acids via hydrogen-bonding interactions), protein domains that dimerize or multimerize with or without the need for a chemical compound/molecule to induce the protein-protein binding (such as DmrA/DmrB/DmrC (Takara Bio), FKBP/FRB..⁵⁵ dDZFs..⁵⁶ and Leucine zippers..⁵⁷), single chain variable fragments..⁵⁸ nanobodies..⁵⁹ affibodies..⁶⁰ proteins that bind to DNA and/or RNA, proteins with quaternary structural interactions, optogenetic protein domains that can dimerize or multimerize in the presence of certain light wavelengths..⁶¹ and/or naturally reconstituting split proteins..⁶²

(172) In some embodiments, the cargo comprises a fusion to a dimerization domain or protein-protein binding domain that may or may not require a molecule to trigger dimerization or protein-protein binding.

(173) In some embodiments, the producer cells are FDA-approved cells lines, allogenic cells, and/or autologous cells derived from a donor.

(174) In some embodiments, the full or active peptide domains of human CD47 may be incorporated in the heVLP surface to reduce immunogenicity.

(175) Examples of AAV proteins included here are AAV REP 52, REP 78, and VP1-3. The capsid site where proteins can be inserted is T138 starting from the VP1 amino acid counting..⁶³ Dimerization domains could be inserted at this point in the capsid, for instance.

(176) Examples of dimerization domains included here that may or may not need a small molecule inducer are dDZF1..⁵⁶ dDZF2..⁵⁶ DmrA (Takara Bio), DmrB (Takara Bio), DmrC (Takara Bio), FKBP..⁵⁵ FRB..⁵⁵ GCN4 scFv..⁵⁸ 10×/24×GCN4..⁵⁸ GFP nanobody..⁵⁹ and GFP..⁶⁴

(177) Examples of split inteins included here are Npu DnaE, Cfa, Vma, and Ssp DnaE..⁵²

(178) Examples of other split proteins included here that make a covalent bond together are Spy Tag and Spy Catcher..⁵³

(179) Examples of RNA binding proteins included here are MS2, Com, and PP7..⁶⁵

(180) Examples of synthetic DNA-binding zinc fingers included here are ZF6/10, ZF8/7, ZF9, MK10, Zinc Finger 268, and Zinc Finger 268/NRE..^{66,67}

(181) Examples of proteins that multimerize as a result of quaternary structure included here are E. coli ferritin, and the other chimeric forms of ferritin..^{68,69}

(182) Examples of optogenetic “light-inducible proteins” included here are Cry2, CIBN, and Lov2-Ja..⁶¹

(183) Examples of peptides the enhance transduction included here are L17E..⁷⁰ Vectofusin-1 (Miltenyi Biotec), KALA..⁷¹ and the various forms of nisin..⁷²

(184) In another embodiment, T1-T4 heVLPs that are produced and isolated can be loaded with biomolecule or chemical molecule cargo by utilizing nucleofection, lipid, polymer, or CaCl.₂ transfection, sonication, freeze thaw, incubation at various temperatures, and/or heat shock of purified particles mixed with cargo. These techniques are adapted from techniques employed to load cargo into exosomes for therapeutic or research applications..⁷³⁻⁷⁵ For example, 100 ug of heVLPs can be resuspended in 200-450 ul of 50 mM trehalose in PBS, mixed with cargo at a desired concentration, and electroporated (GenePulser II Electroporation System with capacitance extender, Bio-Rad, Hercules, CA, USA) in a 0.4 cm cuvette at 0.200 kV and 125 uF.

(185) Production of Cargo-Loaded heVLPs and Compositions

(186) Preferably heVLPs are harvested from cell culture medium supernatant 36-48 hours post-transfection, or when heVLPs are at the maximum concentration in the medium of the producer cells (the producer cells are expelling particles into the media and at some point in time, the particle concentration in the media will be optimal for harvesting the particles). Supernatant can be purified

by any known methods in the art, such as centrifugation, ultracentrifugation, precipitation, ultrafiltration, and/or chromatography. In some embodiments, the supernatant is first filtered, e.g., to remove particles larger than 1 μm , e.g., through 0.45 pore size polyvinylidene fluoride hydrophilic membrane (Millipore Millex-HV) or 0.8 μm pore size mixed cellulose esters hydrophilic membrane (Millipore Millex-AA). After filtration, the supernatant can be further purified and concentrated, e.g., using ultracentrifugation, e.g., at a speed of 80,000 to 100,000 $\times$ g at a temperature between 1° C. and 5° C. for 1 to 2 hours, or at a speed of 8,000 to 15,000 g at a temperature between 1° C. and 5° C. for 10 to 16 hours. After this centrifugation step, the heVLPs are concentrated in the form of a centrifugate (pellet), which can be resuspended to a desired concentration, mixed with transduction-enhancing reagents, subjected to a buffer exchange, or used as is. In some embodiments, heVLP-containing supernatant can be filtered, precipitated, centrifuged and resuspended to a concentrated solution. For example, polyethylene glycol (PEG), e.g., PEG 8000, or antibody-bead conjugates that bind to heVLP surface proteins or membrane components can be used to precipitate particles. Purified particles are stable and can be stored at 4° C. for up to a week or -80° C. for years without losing appreciable activity.

(187) Preferably, heVLPs are resuspended or undergo buffer exchange so that particles are suspended in an appropriate carrier. In some embodiments, buffer exchange can be performed by ultrafiltration (Sartorius Vivaspin 500 MWCO 100,000). An exemplary appropriate carrier for heVLPs to be used for in vitro applications would preferably be a cell culture medium that is suitable for the cells that are to be transduced by heVLPs. Transduction-enhancing reagents that can be mixed into the purified and concentrated heVLP solution for in vitro applications include reagents known by those familiar with the art (Milenyl Biotec Vectofusin-1, Millipore Polybrene, Takara Retronectin, Sigma Protamine Sulfate, and the like). After heVLPs in an appropriate carrier are applied to the cells to be transduced, transduction efficiency can be further increased by centrifugation. Preferably, the plate containing heVLPs applied to cells can be centrifuged at a speed of 1,150 g at room temperature for 30 minutes. After centrifugation, cells are returned into the appropriate cell culture incubator (humidified incubator at 37° C. with 5% CO₂).

(188) An appropriate carrier for heVLPs to be administered to a mammal, especially a human, would preferably be a pharmaceutically acceptable composition. A “pharmaceutically acceptable composition” refers to a non-toxic semisolid, liquid, or aerosolized filler, diluent, encapsulating material, colloidal suspension or formulation auxiliary of any type. Preferably, this composition is suitable for injection. These may be in particular isotonic, sterile, saline solutions (monosodium or disodium phosphate, sodium, potassium, calcium or magnesium chloride and similar solutions or mixtures of such salts), or dry, especially freeze-dried compositions which upon addition, depending on the case, of sterilized water or physiological saline, permit the constitution of injectable solutions. Another appropriate pharmaceutical form would be aerosolized particles for administration by intranasal inhalation or intratracheal intubation.

(189) The pharmaceutical forms suitable for injectable use include sterile aqueous solutions or suspensions. The solution or suspension may comprise additives which are compatible with heVLPs and do not prevent heVLP entry into target cells. In all cases, the form must be sterile and must be fluid to the extent that the form can be administered with a syringe. It must be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms, such as bacteria and fungi. An example of an appropriate solution is a buffer, such as phosphate buffered saline.

(190) Methods of formulating suitable pharmaceutical compositions are known in the art, see, e.g., Remington: The Science and Practice of Pharmacy, 21st ed., 2005; and the books in the series Drugs and the Pharmaceutical Sciences: a Series of Textbooks and Monographs (Dekker, NY). For example, solutions or suspensions used for parenteral, intradermal, or subcutaneous application can include the following components: a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerine, propylene glycol or other synthetic solvents;

antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as ethylenediaminetetraacetic acid; buffers such as acetates, citrates or phosphates and agents for the adjustment of tonicity such as sodium chloride or dextrose. pH can be adjusted with acids or bases, such as hydrochloric acid or sodium hydroxide. The parenteral preparation can be enclosed in ampoules, disposable syringes or multiple dose vials made of glass or plastic.

(191) Pharmaceutical compositions suitable for injectable use can include sterile aqueous solutions (where water soluble) or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersion. For intravenous administration, suitable carriers include physiological saline, bacteriostatic water, Cremophor EL™ (BASF, Parsippany, NJ) or phosphate buffered saline (PBS). In all cases, the composition must be sterile and should be fluid to the extent that easy syringability exists. It should be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), and suitable mixtures thereof. The proper fluidity can be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. Prevention of the action of microorganisms can be achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars, polyalcohols such as mannitol, sorbitol, sodium chloride in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent that delays absorption, for example, aluminum monostearate and gelatin.

(192) Sterile injectable solutions can be prepared by incorporating the active compound in the required amount in an appropriate solvent with one or a combination of ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the active compound into a sterile vehicle, which contains a basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum drying and freeze-drying, which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.

(193) The compositions comprising cargo-loaded heVLPs can be included in a container, pack, or dispenser together with instructions for administration.

EXAMPLES

(194) The invention is further described in the following examples, which do not limit the scope of the invention described in the claims.

(195) Methods

(196) heVLP particles were produced by HEK293T cells using polyethylenimine (PEI) based transfection of plasmids. PEI is Polyethylenimine 25 kD linear (Polysciences #23966-2). To make a stock 'PEI MAX' solution, 1g of PEI was added to 1 L endotoxin-free dH.sub.2O that was previously heated to ~80° C. and cooled to room temperature. This mixture was neutralized to pH 7.1 by addition of TON NaOH and filter sterilized with 0.22 µm polyethersulfone (PES). PEI MAX is stored at -20° C.

(197) HEK293T cells were split to reach a confluency of 70%-90% at time of transfection and are cultured in 10% FBS DMEM media. Cargo vectors, such as one encoding a CMV promoter driving expression of a hPLCδ1 PH fusion to codon optimized Cas9 were co-transfected with a U6 promoter-sgRNA encoding plasmid, a hERVK.sub.conGAG (hGAGK.sub.con) encoding plasmid, and a hENVW (Syncytin-1) encoding plasmid. Transfection reactions were assembled in reduced serum media (Opti-MEM; GIBCO #31985-070). For heVLP particle production on 10 cm plates, 5 µg PH-Cas9 expressing plasmid, 5 µg sgRNA-expression plasmid, 5 µg hERVK.sub.conGAG

expression plasmid, and 5 μ g Syncytin-1 expression plasmid were mixed in 1 mL Opti-MEM, followed by addition of 27.5 μ l PEI MAX. After 20-30 min incubation at room temperature, the transfection reactions were dispersed dropwise over the HEK293T cells.

(198) heVLPs were harvested at 48-72 hours post-transfection. heVLP supernatants were filtered using 0.8 μ m pore size mixed cellulose esters membrane filters and transferred to polypropylene Beckman ultracentrifuge tubes that are used with the SW28 rotor (Beckman Coulter #326823). Each ultracentrifuge tube is filled with heVLP-containing supernatant from 3 10 cm plates to reach an approximate final volume of 35-37.5 ml. heVLP supernatant underwent ultracentrifugation at approximately 100,000 $\times$ g, or 25,000 rpm, at 4° C. for 2 hours. After ultracentrifugation, supernatants were decanted and heVLP pellets resuspended in DMEM 10% FBS media such that they are now approximately 1,000 times more concentrated than they were before ultracentrifugation. heVLPs were added dropwise to cells that were seeded in a 24-well plate 24 hours prior to transduction. Polybrene (5-10 μ g/mL in cell culture medium; Sigma-Aldrich #TR-1003-G) was supplemented to enhance transduction efficiency, if necessary. Vectofusin-1 (10 μ g/mL in cell culture medium, Miltenyi Biotec #130-111-163) was supplemented to enhance transduction efficiency, if necessary. Immediately following the addition of heVLPs, the 24-well plate was centrifuged at 1,150 $\times$ g for 30 min at room temperature to enhance transduction efficiency, if necessary.

Example 1

(199) HEK 293T cells were transduced with T1heVLPs containing PLC PH fused to spCas9, hGAGK.sub.con fused to spCas9, or hArc fused to spCas9 targeted to VEGF site #3. T1heVLPs were pseudotyped with either hENVW (left chart) or hENVFRD (right chart). Gene modification was measured by amplicon sequencing. Particle purification and concentration was performed by PVDF filtration and ultracentrifugation at 100,000 $\times$ g for 2 hours. Results are shown in FIG. 3. Importantly, if HERV-derived GAG (hGAGK.sub.con) was not overexpressed by itself in producer cells, then efficient delivery was not achieved.

(200) TABLE-US-00001 TABLE 1 Position in sequence entry # HERV envelope Gene name Accession no. (a) 1. hENVH1 envH/p62 AJ289709.1 6313-8067 (+) 2. hENVH2 envH/p60 AJ289710.2 5393-7084 (+) 3. hENVH3 envH/p59 AJ289711.1 5204-6871 (+) 4. hENVK1 envK1 AC074261.3 93508-95604 (+) 5. hENVK2 envK2/HML-2.HOM AC072054.10 30365-32464 (-) 6. hENVK3 envK3/C19 Y17833.1 5581-7680 (+) 7. hENVK4 envK4/K109 AF164615.1 6412-8508 (+) 8. hENVK5 envK5/K113 AY037928.1 6451-8550 (+) 9. hENVK6 envK6/K115 AY037929.1 6442-8541 (+) 10. hENVVT envT AC078899.1 154738-156618 (+) 11. hENVW Syncytin-1 AC000064.1 35879-37495 (+) 12. hENVFRD Syncytin-2 AL136139.6 21355-22972 (-) 13. hENVR erv-3 AC073210.8 54963-56978 (-) 14. hENVR(b) envRb AC093488.1 78681-80225 (+) 15. hENVF(c)2 envFc2 AC016222.4 85216-86963 (+) 16. hENVF(c)1 envFc1 AL354685.2 46744-48717 (-) *17. hENVK.sub.con N/A N/A N/A (a) '+' and '-' refer to the orientation within the sequence entry *hENVK.sub.con is a consensus sequence derived from ten proviral ENV sequences. The ENV sequences used to derive this consensus ENV sequence are from the following HERVs: HERV-K113, HERV-K101, HERV-K102, HERV-K104, HERV-K107, HERV-K108, HERV-K109, HERV-K115, HERV-K11p22, and HERV-K12q13.

(201) TABLE-US-00002 TABLE 2 Exemplary Potential Cas9 and Cas12a orthologs DNA-binding Cas ortholog Enzyme class Nickase mutation CI mutations SpCas9 Type II-A D10A D10A, H840A SaCas9 Type II-A D10A D10A, CjCas9 Type II-C D8A D8A, NmeCas9 Type II-C D16A D16A, H588A asCas12a Type II-C D908A, E993A IbCas12a Type II-C D832A, E925A Nickase mutation residues represents a position of the enzyme either known to be required for catalytic activity of the conserved RuvC nuclease domain or predicted to be required for this catalytic activity based on sequence alignment to CjCas9 where structural information is lacking (* indicates which proteins lack sufficient structural information). All positional information refers to the wild-type protein sequences acquired from uniprot.org.

(202) TABLE-US-00003 TABLE 3 Exemplary Deaminase domains and their substrate sequence preferences. Deaminase Nucleotide sequence preference hAID 5'-WRC rAPOBEC1* 5'-TC ≥ CC ≥ AC > GC mAPOBEC3 5'-TYC hAPOBEC3A 5'-TCG hAPOBEC3B 5'-TCR > TCT hAPOBEC3C 5'-WYC hAPOBEC3F 5'-TTC hAPOBEC3G 5'-CCC hAPOBEC3H 5'-TTCA~TTCT~TTCG > ACCCA > TGCA ecTadA hAdar1 hAdar2 Nucleotide positions that are poorly specified or are permissive of two or more nucleotides are annotated according to IUPAC codes, where W = A or T, R = A or G, and Y = C or T.

(203) Nucleotide positions that are poorly specified or are permissive of two or more nucleotides are annotated according to IUPAC codes, where W=A or T, R=A or G, and Y=C or T.

(204) TABLE-US-00004 TABLE 4 Exemplary Epigenetic modulator domains. Epigenetic modulator Epigenetic modulation VP16 transcriptional activation VP64 transcriptional activation P65 transcriptional activation RTA transcriptional activation KRAB transcriptional repression MeCP2 transcriptional repression Tet1 Methylation Dnmt3a Methylation

(205) TABLE-US-00005 TABLE 5 Exemplary CRISPR based RNA-guided RNA binding enzymes RNA-binding Cas ortholog Enzyme class LshCas13a Type-VI LwaCas13a Type-VI PspCas13b Type-VI RfxCas13d Type-VI

(206) TABLE-US-00006 TABLE 6 Plasma membrane recruitment domains described in this work. # Plasma membrane recruitment domain Substitution(s) 1. Pleckstrin homology domain of human phospholipase Cδ1 (hPLCδ1) 2. Pleckstrin homology domain of human Akt1 3. Mutant Pleckstrin homology domain of E17K human Akt1 4. hArc *5. hGAGK.sub.con 6. Pleckstrin homology domain of human 3- phosphoinositide-dependent protein kinase 1 (hPDK1) 7. Human CD9 8. Human CD47 9. Human CD63 10. Human CD81 *hGAGK.sub.con is a consensus sequence derived from ten proviral GAG sequences. The GAG sequences used to derive this consensus GAG sequence are from the following HERVs: HERV-K113, HERV-K101, HERV-K102, HERV-K104, HERV-K107, HERV-K108, HERV-K109, HERV-K115, HERV-K11p22, and HERV-K12q13.

Relevant Protein Sequences:

(207) TABLE-US-00007 *Homo sapiens*: Arc (SEQ ID NO: 1)
MELDHRTSGGLHAYPGPRGGQVAKPNVILQIGKCRAEMLEHVRRT
HRHLLAEVSKQVERELKGLHRSVGKLESNLDGYVPTSDSQRWKKS
IKACLCRCQETIANLERWVKREMHVWREVFYRLERWADRLESTGG
KYPVGSESARHTVSVGVGGPESYCHEADGYDYTVSPYAITPPPA
GELPGQEPAAEQYQPWVPGEDGQPSPGVDTQIFEDPREFLSHLE
EYLRQVGGSEYWLSQIQNHMNGPAKKWWEFKQGSVKNWVEFKKE
FLQYSEGTL SREAIQRELDLPQKQGEPLDQFLWRKRDLYQTLYVD
ADEEEIIQYVVGTLQPKLKRFLRHPLPKTLEQLIQRGMEVQDDLE
QAAEPAGPHLPVEDEAETLTPAPNSESVASDRTQPE >AJ289709.1 Human endogenous
retrovirus H HERV-H/env62 HERV_H/ENV_62-hENVH1: (SEQ ID NO: 2)
MIFAGKAPSNTSTLMKFYSLLLYSLLFSFPFLCHPLPLPSYLHHT
INLTHSLLAASNPSLVNNCWLCISLSSSAYTAVPAVQTDWATSPI
SLHLRTSFNSPHLYPPEELIYFLDRSSKTSPDISHQQAALLRTY
LKNLSPYINSTPPIFGPLTTQTTPVAAPLCISWQRPTGIPLGNL
SPSRCSFTLHLRSPTTNINETIGAFQLHITDKPSINTDKLKNISS
NYCLGRHLPCISLHPWLSSPCSSDSPRPSSCLLIPSPENNSERL
LVDTRRFLIHENRTFPSTQLPHQSPLQPLTAAALAGSLGVWVQD
TPFSTPSHLFTLHLQFCLAQGLFFLCGSSTYMCLPANWTGTCTLV
FLTPKIQFANGTEELPVPLMTPTQQKRVIPLIPLMVGLGLSASTV
ALGTGIAGISTSVMTFRSLSNDFSASITDISQTL SVLQAQVDSL A
AVVLQNRRGLDLLTAEKGGLCIFLNEECFYLNQSGLVYDNIKKL
KDRAQKLANQASNYAEPWALS NWMSWVLP IVSPLIPIFLLLLFG
PCIFRLVSQFIQNRIQAITNHSIRQMFLLTSPQYHPLPQDLPSA >AJ289710.2 Human

endogenous H HERV-H/env60-HERV_H_ENV_60-hENVH2: (SEQ ID NO: 3) MIFAGRASSNTSTLMKFYSLLLYSLLFSFPILCHPLPLPSYLHHT
INLTHSLLAVSNPSLAKNCWLCISLPSSAYPAVPALQTDWGTSPV
SPHLRTSFNSPHLYPPEKLIYFLDRSSKTSPDISHQQAALLCTY
LKNLSPYINSTPPTFGPLTTQTTIPVAAPLCISRQRPTGIPLGNL
SPSRCSFTLHLRSPTHITETNGAFQLHITDKPSINTDKLKNVSS
NYCLGRHLSCISLHPWLFSPCSDSPRPSSCLLIPSPKNNSESL
LVDAQRFLLIYHENRTSPSTQLPHQSPLQPLTAAPLGGSLRVWVQD
TPFSTPSHLFTLHLQFCLVQSLFFLCGSSTYMCLPANWTGTCTLV
FLTSKIQFANGTEELPVPLMTPTTRQKRVIPLIPLMVGLGLSASTV
ALGTGIAGISTSVTTFRILSNDFSASITDISQTL SGLQAQVDSSA
AVVLQNRQGLDLLTAEKGGLCIFLNEESYFYLNQSGLVYDNIKKL
KDKAQNLANQASNYAEPWPLSNWMSWVLPILSPLIPIFLLLFFR
PCIFHLVSQFIQNHIAITDHSI >AJ289711.1 Human endogenous retrovirus H HERV-
H/env59-HERV_H_ENV_59-hENVH3: (SEQ ID NO: 4)

MILAGRAPSNTSTLMKFYSLLLYSLLFSFPFLYHPLPLPSYLHHT
INLTHSLPAASNPSLANNCWLCISLSSSAYIAVPTLQTD RATSPV
SLHLRTSFNSPHLYPPEELIYFLDRSSKTSPDISHQPAAALLHIY
LKNLSPYINSTPIFGPLTTQTTIPVAAPLCISRQRPTGIPLGNI
SPSRCSFTLHLQSPTHVTETIGVFQLHIIDKPSINTDKLKNVSS
NYCLGRHLPYISLHPWLPSPCSDSPRPSSCLLTPSPQNNSERL
LVDTQRFLIHENRTSSSMQLAHQSPLQPLTAAALAGSLGVWVQD
TPFSTPSHPFSLHLQFCLTQGLFFLCGSSTYMCLPANWTGTCTLV
FLTPKIQFANGTKELPVPLMTLTPQKRVIPLIPLMVGLGLSASTI
ALSTGIAGISTSVTTFRSPSNDFSASITDISQTL SVLQAQVDSL A
AVVLQNRRLGLGLSILLNEECFYLNQSGLVYENIKKLKDRAQKLA
NQASNYAESPWALS NWMSWVLPILSPLIPIFLLLLFGPCIFHLVS QFIQNRIQAITNHSI
>AC074261.3 *Homo sapiens* chromosome 12 clone RP11-55F19 envK1- ENVK1:
(SEQ ID NO: 5) MHPSEMQRKAPPRRRRHRNRAPLTHKMNMVMTSEQMKLPSTKKA
PPTWAQLKKLTQLATKYLENTKV TQTPESMLLAALMIVSMVVS L P
MPAGAAAANYTNWAYVPFPPLIRAVTWMDNPIEVYVND SVWVHGP
IDDRCPAKPEEEGMMINISIGYHYPPICLGRAPGCLMPAVQNWLV
EVPTVSPISRFTYNMVSGMSLRPRVNYLQDFS YQRS LKFRPKGK P
CPKEIPKESKNTEVLVWEECVANSVVILQNNEFGTIIDWAPRGQF
YHNCSGQTQSCPSAQVSPA VDSDLTESLDKHKHKKLQSFYPWEWG
EKGISTPRPKIISPVSGPEHPELWRLTVASHHIRIWSGNQTLETR
DRKPFYTVDLNSSLTVP LQSCVKPPYMLVVGNIVIKPDSQTITCE
NCRL LTCIDSTFNWQHRILLVRAREGVWIPVSMDRPWEASPSIHI
LTEVLKGV LNRSKRFIFTLIAVIMGLIAVTAMAAVAGVALHSFVQ
SVN FVNDWQKNSTR L WNSQSSIDQKLANQINDLRQTVIWMGDRLM
SLEHRFQLQCDWNTSDFCITPQIYNESEHHWDMVRRHLQGREDNL
TLDISKLKEQIFEASKAHLNLVPGTEA IAGVADGLANLNPVTWVK
TIGSTTIINLILVCLFCLLLVCRFTQQLRRDSYHRERAMMTMV
VLSKRKGGNVGKSKRDQIVTVSV >AC072054.10 *Homo sapiens* BAC clone RP11-
33P21-ENVK2: (SEQ ID NO: 6)

MNPSEMQRKAPPRRRRHRNRAPLTHKMNMVMTSEEQMKLPSTKKA
EPPTWAQLKKLTQLATKYLENTKV TQTPESMLLAALMIVSMVVS L
PMPAGAAAANYTYWAYVPFPPLIRAVTWMDNPTEVYVND SVWVPG
PIDDRCPAKPEEEGMMINISIGYHYPPICLGRAPGCLMPAVQNWLV
VEVPTVSPICRFTYH MVSGMSLRPRVNYLQDFS YQRS LKFRPKGK

PCPKEIPKESKNTTEVLVWEECVANSVILQNNNEFGTIIDWAPRGQ
FYHNCSGQTQSCPSAQVSPAVIDSDLTESLDKHKHKKLQSFYPWEW
GEKGISTPRPKIVSPVSGPEHPELWRLTVASHHIRIWSGNQTTLET
RDRKPFYITIDLNSSLTVPQLQSCVKPPYMLVVGNIVIKPDSQTITC
ENCRLTLCIDSTFNWQHRILLVRAREGVWIPVSMRDPWEASPSVH
ILTEVLKGVNLNRSKRFIFTLIAVIMGLIAVTATAAVAGVALHSSV
QSVNFVNDWQKNSTRLWNSQSSIDQKLANQINDLRQTVIWMGDRL
MSLEHRFQLQCDWNTSDFCITPQIYNESEHHWDMVRRHLQGREDN
LTLDISKLKEQIFEASKAHLNLVPGTEAIAGVADGLANLNPVTWV
KTIGSTTIINLILILVCLFCLLLVCRCTQQLRRDSHDRERAMMTM
AVLSKRKGGNVGKSKRDQIVTVSV >Y17833.1 Human endogenous retrovirus K
(HERV-K) envK3-ENVK3: (SEQ ID NO: 7)
MNPSEMQRKAPRRRRHRNRAPLTHKMNMVMTSEEQMKLPSTKKA
EPPTWAQLKKLTQLATKYLENTKVTQTPESMLLAALMIVSMVVSL
PMPAGAAAANYTYWAYVPFPPLIRAVTWMDNPIEVYVNDVWVPG
PTDDHCPAKPEEEGMMINISIGYRYPICLGRAPGCLMPAVQNW
VEVPTVSPISRFTYHMOVSGMSLRPRVNYLQDFSYQRSFKFRPKGK
PCPKEIPKESKNTTEVLVWEECVANSVILQNNNEFGTIIDWAPRGQ
FYHNCSGQTQSCPSAQVSPAVIDSDLTESLDKHKHKKLQSFYPWEW
GEKGISTPRPKIISPVSPEHPELWRLTVASHHIRIWSGNQTTLET
RDRKPFYITVDLNSSVTVPQLQSCIKPPYMLVVGNIVIKPDSQTITC
ENCRLTLCIDSTFNWQHRILLVRAREGVWIPVSMRDPWETSPSIH
TLTEVLKGVNLNRSKRFIFTLIAVIMGLIAVTATAAVAGVALHSSV
QSVNFVNDWQKNSTRLWNSQSSIDQKLANQINDLRQTVIWMGDRL
MSLEHRFQLQCDWNTSDFSITPQIYNESEHHWDMVRRHLQGREDN
LTLDISKLKEQIFEASKAHLNLVPGTEAIAGVADGLANLNPVTWV
KTIGSTTIINLILILVCLFCLLLVCRCTQQLRRDSHDRERAMMTM
AVLSKRKGGNVGKSKRDQIVTVSV >AF164615.1 Homo sapiens endogenous retrovirus
HERV-K109 envK4-ENVK4: (SEQ ID NO: 8)
MNPSEMQRKAPRRRRHRNRAPLTHKMNMVMTSEEQMKLPSTKKA
EPPTWAQLKKLTQLATKYLENTKVTQTPESMLLAALMIVSMVVSL
PMPAGAAAANYTNWAYVPFPPLIRAVTWMDNPIEVYVNDVWVPG
PIDDRCPAKPEEEGMMINISIGYRYPICLGRAPGCLMPAVQNWLV
EVPIVSPICRFTYHMOVSGMSLRPRVNYLQDFSYQRSFKFRPKGK
CPKEIPKESKNTTEVLVWEECVANSVILQNNNEFGTIIDWTPQGQF
YHNCSGQTQSCPSAQVSPAVIDSDLTESLDKHKHKKLQSFYPWEWG
EKGISTPRPKIISPVSPEHPELWRLTVASHHIRIWSGNQTTLET
RDRKPFYITVDLNSSLTLPQLQSCVKPPYMLVVGNIVIKPDSQTITCE
NCRLTLCIDSTFNWQHRILLVRAREGVWIPVSMRDPWEASPSIHI
LTEVLKGVNLNRSKRFIFTLIAVIMGLIAVTATAAVAGVALHSSVQ
SVNFVNDGQKNSTRLWNSQSSIDQKLANQINDLRQTVIWMGDRLM
SLEHRFQLQCDWNTSDFCITPQIYNESEHHWDMVRRHLQGREDNL
TLDISKLKEQIFEASKAHLNLVPGTEAIAGVADGLANLNPVTWVK
TIGSTTIINLILILVCLFCLLLVCRCTQQLRRDSHDRERAMMTMA
VLSKRKGGNVGKSKRDQIVTVSV >AY037928.1 Human endogenous retrovirus K113
envK5-ENVK5: (SEQ ID NO: 9)
MNPSEMQRKAPRRRRHRNRAPLTHKMNMVMTSEEQMKLPSTKKA
EPPTWAQLKKLTQLATKYLENTKVTQTPESMLLAALMIVSMVVSL
PMPAGAAAANYTYWAYVPFPPLIRAVTWMDNPIEIVVNDVWVPG
PTDDCCPAKPEEEGMMINISIGYRYPICLGRAPGCLMPAVQNW

VEVPTVSPISRFTYHMSVSGMSLRPRVNYLQDFSYQSRSLKFRPKGK
PCPKEIPKESKNTEVLVWEECVANS AVILQNNEFGTLIDWAPRGQ
FYHNCSGQTQSCPSAQVSPA VDSDLTESLDKHKHKKLQSFYPWEW
GEKGISTARPKIISPVSGPEHPELWRLTVASHHIRIWSGNQTTLET
RDRKPFYTTIDLNSSLTVP LQSCVKPPYMLVVGNIVIKPDSQTITC
ENCRL LTCIDSTFNWQHRILLVRAREGVWIPVSM DRPWEASPSVH
ILTEVLKGV LNRSKRFIFTLIAVIMGLIAVTATAAVAGVALHSSV
QSVNFVNDWQNNSTR LWNSQSSIDQKLANQINDLRQT VIWMGDRL
MSLEHRFQLQCDWNTSDFCITPQIYNESEHHWDMVRCHLQGREDN
LTLDISK LKEQIFEASKAHLNLVPGTEAIAGVADGLANLNTVTWV
KTIGSTTIINLILILVCLFCLLLVYRCTQQLRRDS DHRERAMMTM
VVLSKRKGGNVGKSKRDQIVTVSV >AY037929.1 Human endogenous retrovirus
K115 envK6-ENVK6: (SEQ ID NO: 10)

MNPSEMQRKAPRRRRHRNRAPLTHKMNMVTSEEQMKLPSTKKA
EPPTWAQLKKLTQLATKYLENTKVTQTPE SMLLAALMIVSMVVSL
PMPAGAAVANYTNWAYVPFPPLIRAVTWMDNPIEVYVND SVWVPG
PIDDRCPAKPEEEGMMINISIGYRYPPICLGRAPGCLMPAVQNW L
VEVPTVSPISRFTYHMSVSGMSLRPRVNYLQDFSYQSRSLKFRPKGK
PCPKEIPKESKNTEVLVWEECVANS AVILQNNEFGTIIDWAPRGQ
FYHNCSGQTQSCPSAQVSPA VDSDLTESLDKHKHKKLQSFYPWEW
GEKRISTPRPKIVSPVSGPEHPELWRLTVASHHIRIWSGNQTTLET
RDRKPFYTVDLNSSLTLP LQSCVKPPYMLVVGNIVIKPDSQTITC
ENCRL LTCIDSTFNWQHRILLVRAREGVWIPVSM DRPWEASPSVH
ILTEVLKGV LNRSKRFIFTLIAVIMGLIAVTATAAVAGVALHSSV
QSVNFVNDGQKNSTR LWNSQSSIDQKLANQINDLRQT VIWMGDRL
MSLEHRFQLQCDWNTSDFCITPQIYNDSEHHWDMVRRHLQGREDN
LTLDISK LKEQIFEASKAHLNLVPGTEAIAGVADGLANLNPVTWV
KTIGSTTIINLILILVCLFCLLLVCRCTQQLRRDS DHRERAMMTM
AVLSKRKGGNVGKSKRDQIVTVSV >AC078899.1 Homo sapiens chromosome 19,
BAC BC371065 envT-ENV T: (SEQ ID NO: 11)

MGPEAWVRPLKTAPKPGEAIRLILFIYLS CFFLPVMSSEPSYSFL
LTSFTTGRVFANTTW RAGTSKEVSFAVDLCVLFPEPARTHEEQHN
LPVIGAGSVDLAAGFGHSGSQTGCGSSKGA EKGLQNVDFYLCPGN
HPDASCRDTYQFFCPDWTCVTLATYSGGSTRSSTLSISR VPHPKL
CTRKN CNPLTITVHDPNAAQWYYGMSWGLRLYIPGFDVGT MFTIQ
KKILVSWSSPKPIGPLTDLGDPIFQKH PDKVDLTVPLPFLVPRPQ
LQQQHLQPSLMSILGGVHHLLNLTQPKLAQDCWLCLKAKPPYYVG
LGVEATLKRGPLSCHTRPRALTIGDVSGNASCLISTGYNLSASPF
QATCNQSL LTSISTSVSYQAPNNTWLACTSGLTRCINGTEPGPLL
CVLVHVLPQVYVYSGPEGRQLIAPPELHPRLHQAVPLL VPLL AGL
SIAGSAAIGTAALVQGETGLISLSQQVDADFSNLQSAIDILHSQV
ESLAEVVLQNCRC LDLLFLSQGGLCAALGESCCFYANQSGVIKGT
VKKVRENLD RHQQERENNIPWYQSMFNWNPWLTTLITGLAGPLLI
LLLSLIFGPCILNSFLNFIKQRIASVKLT YLKTQYDTLVNN >AC000064.1 Human BAC
clone RG083M05 from 7q21-7q22 envW (Syncytin-1)-ENVW (Syncytin-1): (SEQ
ID NO: 12) MALPYHIFLFTVLLPSFTLTAPPPCRCMTSSSPYQEFLWRMQRPG
NIDAPSYRSLSKGTPTFTAHTHMPRNCYHSATLCMHANTHYWTGK
MINPSCPGGLGVTVCW TYFTQTGMSDGGGVQDQAREKHVKEVISQ
LTRVHGTSSPYKGLDLSKLHETLRTHTRLVSLFNTTTLTGLHEVSA
QNPTNCWICLPLNFRPYVSIPVPEQWNNFSTEINTTSVLVGPLVS

NLEIHTSNTLTHCTNSLTKVNTQVCLPSGIF
VCGTSAYRCLNGSSESMCFLSFLVPPMTIYTEQDLYSYVISKPRN
KRVPILPFVIGAGVLGALGTGIGGITTSTQFYKLSQELNGDMER
VADSLVTLQDQLNSLA AVVLQNRRLDLLTAERGGTCLFLGEECC
YYVNQSGIVTEKVKEIRDRIQRRAEELRNTGPWGLLSQWMPWILP
FLGPLAAIILLLLFGPCIFNLLVNFVSSRIEAVKLQMEPKMQSKT
KIYRRPLDRPASPRSDVNDIKGTPPEEISAAQPLLRPNSAGSS >AL136139.6 Human
DNA sequence from clone RP4-76112 envFRD-ENVFRD (Syncytin-2): (SEQ ID
NO: 13) MGLLLLVLILTPSLAAYRHPDFPLLEKAQQLLQSTGSPYSTNCWL
CTSSSTETPGTAYPASPREWTSIEAELHISYRWDPNLKGLMRPAN
SLLSTVKQDFDIRQKPPIFGPIFTNINLMGIAPICVMAKRKNGT
NVGTL PSTVCNVTFTVDSNQQTYQTYTHNQFRHQPRFPKPPNITF
PQGTLLDKSSRFCQGRPSSCSTRNFWFRPADYNQCLQISNLSSTA
EWVLLDQTRNSLFWENKTKGANQSQTPCVQVLAGMTIATS YLGIS
AVSEFFGTSLTPLFHFHISTCLKTQGA FYICGQSIHQCLPSNWTG
TCTIGYVTPDIFIAPGNLSLPIPIYGNSPLPRVRAIHFIPLLAG
LGILAGTGTGIAGITKASLTYSQLSKEIANNIDTMAKALTTMQEQ
IDSLAAVVLQNRRLDMLTAAQGGICLALDEKCCFWVNQSGKVQD
NIRQLLNQASSLRERATQGWLNWEGTWKWF SWVLPLTGPLVSLLL
LLLFGPCLLNLITQFVSSRLQAIKLQTNLSAGRHP RNIQESPF >AC073210.8 Homo
sapiens BAC clone RP11-460N20 envR-ENVR: (SEQ ID NO: 14)
MLGMNMLLITLFLLLPLSMLKGEPWEGCLHCTHTTWSGNIMTKTL
LYHTYYECAGTCLGTCTHNQTTYSVCDPGRGQPYVCYDPKSSPGT
WFEIHVGSKEGDLLNQTKVFP SGKD VVS LYFDVCQIVSMGSLFPV
IFSSMEYYSSCHKNR YAHPACSTDSPVTT CWDCTTWSTNQQLGP
IMLT KIPLEPDCKTSTCNSVNL TILEPDQPIWTTGLKAPLGARVS
GEEIGPGAYVYLYIIKKTRTRSTQQFRVFESFYEHVNQKLPEPPP
LASNLFAQLAENIASSLHVASCYVCGGMNMGDQWPWEARELMPQD
NFTLTASSLEPAPSSQSIWFLKTSIIGKFCIARWGKAFTDPVGEL
TCLGQQYYNETLGKTLWRGKSNNSESPHPSPF SRFPSLNH SWYQL
EAPNTWQAPSGLYWICGPQAYRQLPAKW SGACVLGTIRPSFFLMP
LKQGEALGYPIYDETKRKSKRGITIGDWKDNEWPPERIIQYYGPA
TWAEDGMWGYRTPVYMLNRI RLQAVLEIITNETAGALNLLAQQA
TKMRNVIIYQNRLALDYLLAQEEGVCGKFNL TNCCLELDDEGKVIK
EITAKIQKLAHIPVQTWKG >AC093488.1 Homo sapiens chromosome 3 clone RP11-
1008 envR(b)- ENVR(b): (SEQ ID NO: 15)
MDPLHTIEKVPARRNIHDRGHQGH RMGDGTPGRPKISVQQMTRFS
LIIFFLSAPFVNASTSNVFLQWAHSYADGLQQGDPCWVCGSLPV
TNTMELPWWV SPLQGKD WVFFQSFIGDLKQWTGAQMTGVTRKNIS
EWPINKTLNEPGHDKPFSVNETRDKVIAFAIPLLDTKVFVQTSRP
QNTQYRNGFLQIWDGFIWLTATKGHLSQIAPLCWEQRNHSLDNWP
NTTRVMGWIPP GQCRHTILLQQRDLFATDWSQQPGLN WYAPNGTQ
WLCSPNLWPWLPSGWLGCCTLGIPWAQGRWVKTMEVYPYLPHVVN
QGTRAIVHRNDHLPTIFMP SVGLGTVIQHIEALANFTQRALNDSL
QSISLMNAEVYYMHEDILQNRMALDILTAAEGGT CALIKTECCVY
IPNNSRNISLALEDTCRQIQVISSALS LHDWIASQFSGRPSWWQ
KILIVLATLWSVGIALCCGLYFCRMFSQH IPQTHSIIFFQQELPLS PPSQEHYQSQRDIFHSNAP
>AC016222.4 Homo sapiens clone RP11-26J6 envF(c)2-ENVF(c)2: (SEQ ID NO:
16) MNSPCDRLQQFIQVLLEESWSFPSFANTLHWPENLLSYIDELVWQ
GSLQNFHQHEVRFDKPPLRLPLTGFS SLTENWSSRQAVSSRLVAT

ASPAGQAPALCFLYDQSNKCN
WVKENVGCPWHWCNIHEALIRTEKGS
DPMFYVNTSTGGRDGFNGF
NLQISDPWDPRWASGVDGGLYEHKTF
MYPVAKIRIARTLKTTVTG
LSDLA
SIQSAEKELTSQ
LQPAADQAKSSRFSWLT
LISEGAQLLQ
STGVQNL
SHCFLCAALRRPPLVAVPLPTPFNYTINSSTPIPPVPK
GQVPLFSDPIRHKFPFCYSTPNASWCNQTRMLTSTPAPPRGYFWC
NSTLTKVLNSTGNHTLCLPISLIPGLTLYSQDEL
SHLLAWTEPRP
QNKSKWAIFLPLVLGISLASSLVASGLGKGAL
THSIQTSQDLSTH
LQLAIEASAESLDSLQRQITTVAQVAAQNRQALD
LLMAEKGR
TCLFLQEECCYYL
NESGVVENS
LQTLKKKKSSKRS >AL354685.17 Human DNA sequence
from clone RP13-75G22 envF(c)1- ENVF(c)1: (SEQ ID NO: 17)
MARPSPLCLLLLLLLTLLTPIVPSNSLLTEPPFRWRFY
LHETWTQGN
RLSTVTLATVDCQPHGCQAQVTFNFTSFKSVLRGWSNPTICFVYD
QTHSNCRDYWVD
TNGGCPYAYCRMHV
TQLHTAKKLQHTYRLTSDG
RTTYFLTIPDPWDSRWVSGVTGRLYRWPTDSYPVGKLRIFLT
YIRVIPQVLSNLKDQADNIKHQEEVINTLVQSHPKADMV
TYDDKAEAG
PFSWITLVRHGARLVN
MAGLVNLSHCFLCTALSQPP
LVAVPLPQAFNTSGNHTAHPSGVFSE
QVPLFRDPLQPQFPFCYTT
PNSSWCNQTYSGSLSNLSAPAGGYFWC
NFTLTKHLNISSNNTLSRNLCLPISLV
PRLTLYSEAE
LSSLVNPPMRQKRAVFPPLVIGVSLTSSLVASGLG
TGAIVHFISSQDLSIKLQMAIEASAESLASLQRQITSVAKVAMQ
NRRALDLLTADKGGTCMFLGEECCYYINESGLVETSLLTLDKIRD
GLHRPSSTPNYGGGWWQSPLTTWIIPFISPILII
CLLLLLIAPCVL
KFIKNRISEVSRVTVNQMLLHPYSRLPTSEDHYDDALTQ
QEAAR HERV-Kcon ENV-
hENVKcon: (SEQ ID NO: 18)
MNPSEMQRKAPPRRRRRHRNRAPLTHKMNKMVTSEEQMKLPSTKKA
EPPTWAQLKKLTQLATKYLENTKVTQTPE
SMLLAALMIVSMVVSL
PMPAGAAAANYTYWAYVPFPPLIRAVTWMDNPIE
VYVND
SVWVPGPIDDRCPAKPEEEGMMINISIGYRYPPICLGRAPGCLMPAVQ
NWLVEVPTVSPISRFTYH
MVSGMSLRPRVNYLQDFSYQ
RSLKFRPKGKPCPKEIPKESKNT
EVLVWEECVANS
AVILQNN
EFGTIIDWAPRGQFYHNC
SGQTQSCPSAQVSPA
VDSDLTESLDKHKHKKLQSFYPWEW
GEKGISTPRPKIVSPVSGPEHPELWRLTVASHHIRIWSGNQTLET
RDRKPFYTVDLNSSLTVPLQSCVKPPYMLVVGNIVIKPDSQTITC
ENCRL
LTCIDSTFNWQHRILLVRAREGVWIPVSM
DRPWEASPSVHILTEVLKGV
LNRSKRFIFTLIAVIMGLIAVTATAAVAGVALHSSV
QSVN
FVNDWQKNSTR
LWNSQSSIDQKLANQINDLRQT
VIWMGDRLMSLEH
RFQLQCDWNTSDFCITPQIYNESEHHWDMVRRHLQ
GREDNLTLDISKLKEQIFEASKAHLNLVPGTEA
IAGVADGLANLNPVTWV
KTIGSTTIINLILILVCLFCLLLVCRCTQQLRRDS
DHRERAMMTM
AVLSKRKGGNVGKSKRDQIVTVSV HERV-Kcon GAG-hGAGKcon: (SEQ ID NO: 19)
MGQTKSKIKSKYASYLSFIKILLKRGGVKVSTKNLIKLFQII
EQFCPWFPEQGTLDLKD
WKRIGKELKQAGRKGNIPLTVWNDWAIKA
ALEPFQTEEDSVSVSDAPGSCIIDCNENTRKK
SQKETEG
LHCEYVAEPVMAQSTQNVDYNQLQEVIYPETL
KLEGKGPELVGPSESKPRG
TSPLPAGQVPVTLQPQKQVKENKTQPPVAYQY
WPPAELQYRPPPE
SQYGYPGMPAPQGRAPYPQPPTRRLNPTAPPSRQ
GSELHEIIDK
SRKEGDTEAWQFPVTLEPMPPGEGAQEGEPPTVE
ARYKSFSIKML
KDMKEGVKQYGPNSPYMRTL
LDSIAHGHRLIPYDWEILAKSSSLSP
SQFLQFKTWWIDGVQE
QVRRNRAANPPVNIDADQLLGIGQNWSTI

SQQALMQNEALQVRAIQDPGSTCPSFNTVRQGSKEP
YPDFVARLQDVAQKSIADKARKVIVELMAYENANPECQSAIKPL
KGKVPAGSDVISEYVKACDGIGGAMHKAMLMAQAITGVVLGGQVR
TFGGKCYNCGQIGHLLKKNCPVLNKQNITIQATTTGREPPDLCPRC
KKGKHWASQCRSKFDKNGQPLSGNEQRGQPQAPQQTGAFPIQPFV
PQGFQGGQPPLSQVFQGISQLPQYNNCPPPQAAVQQ *Rattus norvegicus* & synthetic:

APOBEC1-XTEN L8-nspCas9-UGI-SV40 NLS (SEQ ID NO: 20)
MSSETGPVAVDPTLRRRIEPHEFEVFFDPRELKRETCCLLYEINWG
GRHSIWRHTSQNTNKHVEVNFIEKFTTERYFCPNTRCSITWFLSW
SPCGECSRAITEFLSRYPHVTLFIYIARLYHHADPRNRQGLRDLI
SSGVTIQIMTEQESGYCWRNFVNYSNEAHWPYPHYLWVRLYVL
ELYCIILGLPPCLNILRRKQPQLTFFTIALQSCHYQRLPPHILWA
TGLKSGSETPGTSESATPESDKKYSIGLAIGTNSVGWAVITDEYK
VPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRR
YTRRKNRICYLQEIFSNEMAKVDDSFHRLEESFLVEEDKKHERH
PIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHM
IKFRGHFLIEGDLNPDNSDVKLFIQLVQTYNQLFEENPINASGV
DAKAILSARLSKSRLENLIAQLPGEKKNGLFGNLIALSLGLTPN
FKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYADLFLAAKNL
SDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQ
QLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGT
EELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYF
FLKDNREKIEKILTRIPYYVGPLARGNSRFAWMTRKSEETITPW
NFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLYEYFTVY
NELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKE
DYFKKIECFDSVEISGVEDRFNASLGTYHDLLKIIKDKDFLDNEE
NEDILEDIVLTTLTFEDREMIEERLKYAHLFDDKVMKQLKRRRY
TGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFMQLIHDDS
LTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIKKGILQTVKVVDL
LVKVMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEELIKEL
GSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQELDINRLSDY
DVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMN
YWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQ
ITKHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQ
FYKVRINNYHHAHDAYLNAVVGTAIIKKYPKLESEFVYGDYKVY
DVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRP
LIETNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFS
KESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVAKVEK
GKSKKLKSVKELLGITIMERSSEFKNPIDFLEAKGYKEVKKDLII
KLPKYSLEFENGRKRMLASAGELQKGNELALPSKYVNFLYLASH
YEKLKGSPEDEQKQLFVEQHKHYLDEIIEQISEFSKRVLADAN
LDKVL SAYNKHDKPIREQAENIHLFTLTNLGAPAAFKYFDTTI
DRKRYTSTKEVLDATLIHQSTGLYETRIDLSQLGGDSGGSTNLS
DIEKETGKQLVIQESILMLPEEVEEVIGNKPESDILVHTAYDES
TDENVMLLTSDAPEYKPWALVIQDSNGENKIKMLSGGSPKKKRKV *Streptococcus*
pyogenes: spCas9 Bipartite NLS (SEQ ID NO: 21)
MDKKYSIGLDIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKK
NLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNE
MAKVDDSFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTI
YHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDLNPDNSD

VDKLFQVLQVQTYNQLFEENQVSDAKAILSARLSKSRRLNL
IAQLPGEKKNGLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKDT
YDDDLNLLAQIGDQYADLFLAAKNLSDAILLSDILRVNTEITKA
PLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYA
GYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTF
DNGSIPHQIHLGELHAILRRQEDFYFPLKDNREKIEKILTRIPY
YVGPLARGNSRFAWMTRKSEETITPWNFEVVDKGASAQSFIERM
TNFDKNLPNEKVLPHSLLYEYFTVYNELTKVKYVTEGMRKPAFL
SGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVED
RFNASLGTYHDLKKIKDKDFLDNEENEDILEDIVLTLTLFEDRE
MIEERLKTIAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQS
GKTILDFLKSDGFANRNFMQLIHDDSLTFKEDIQKAQVSGQGDSL
HEHIANLAGSPAIIKKGILQTVKVVDDELVKVMGRHKPENIVIEMAR
ENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEK
LYLYYLQNGRDMYVDQELDINRLSDYDVDHIVPQSFLKDDSIDNK
VLTRSDKNRGKSDNVPSEEVVKMKMKNYWRQLLNAKLITQRKFDNL
TKAERGGLSELDKAGFIKRLVETRQITKHVAQILDSRMNTKYDE
NDKLIREVKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLN
AVVGTAIIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIGKATAK
YFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDF
ATVRKVLSPQVNIKKTEVQTGGFSKESILPKRNSDKLIARKKD
WDPKKYGGFDSPTVAYSVLVVAKEKKGKSKKLKSVKELLGITIME
RSSFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLELENGRKRMLA
SAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDNEQKQLFVE
QHKHYLDEIIEQISEFSKRVLADANLDKVLSAYNKHRRDKPIREQ
AENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDATLIHQ
SITGLYETRIDLSQLGGDGSGGGGSGKRTADGSEFEPKKKRKVSS
GGDYKDHDGDYKDHDIDYKDDDDDK *Staphylococcus aureus*: saCas9 (SEQ ID NO:
22) MKRNYILGLDIGITSVGYGIIIDYETRDVIDAGVRLFKEANVENNE
GRRSKRGARRLKRRRRRHRIQRVKKLLFDYNLLTDHSELSGINPYE
ARVKGLSQKLSEEEFSAALLHLAKRRGVHNVNEVEEDTGNELSTK
EQISRNSKALEEKYVAELQLERLKKDGEVRGSINRFKTS DYVKEA
KQLLKVQKAYHQLDQSFIDTYIDLLETRRTYYEGPGEGSPFGWKD
IKEWYEMLMGHCTYFPEELRSVKYAYNADLYNALNDLNNLVITRD
ENEKLEYEYEFQIIENVFKQKKKPTLKQIAKEILVNEEDIKGYRV
TSTGKPEFTNLKVYHDIKDITARKEIIEAELLDQIAKILTIYQS
SEDIQEELTNLNSLTQEEIEQISNLKGYTGTHNLSLKAINLILD
ELWHTNDNQIAIFNRLKLVPKKVDLSQQKEIPTTLVDDFILSPVV
KRSFIQSIKVINAIIKKYGLPNDIIIELAREKNSKDAQKMINEMQ
KRNQRQTNERIEEIIRTTGKENAKYLIEKIKLHDMQEGKCLYSLEA
IPLEDLLNNPFNYEVDHII PRSVSFDNSFNKVLVKQEENSKKGN
RTPFQYLSSSDSKISYETFKKHILNLA KGKGRISKTKKEYLLEER
DINRFSVQKDFINRNLVDTRYATRGLMNLLRSYFRVNNLDVKVKS
INGGFTSFLRRKWKFKKERNKGYKHHAE DALIIANADFIFKEWKK
LDKAKKVMENQMFEKQAESMPEIETE QEYKEIFITPHQIKHIKD
FKDYKYSHRVDKKNRELINDTLYSTRKDDKGNTLIVNNLNGLYD
KDNDKLKLINKSPEKLLMYHHDPQTYQKLKLIMEQYGDEKNPLY
KYYEETGNYLTKYSKKDNGPVIKKIKYYGNKLNAHLDITDDYPNS
RNKWVKLSLKPYRFDVYLDNGVYKFVTVKNL DVIKKENYYEVNSK
CYEEAKKLLKKISNQA EFIASFYNNDLIKINGELYRVIGVNNDLLN

RIEVDNIDTYREYLENMDKRRPPIIKTIASKTQSIKYSTDIL GNLYEVKSKKHPQIIKKG
Acidaminococcus sp.: asCas12a (SEQ ID NO: 23)
MTQFEGFTNLYQVSKTLRFELIPQGKTLKHIQEQGFIEEDKARND
HYKELKPIIDRIYKTYADQCLQLVQLDWENLSAAIDSYRKEKTEE
TRNALIEEQATYRNAIHDYFIGRTDNLTDAINKRHAIEIYKGLFKA
ELFNGKVLKQLGTVTTTEHENALLRSFDKFTTYFSGFYENRKNVF
SAEDISTAIPHRIVQDNFPKFKENCHIFTRLITAVPSLREHFENV
KKAIGIFVSTSIEEVFSFPFYNQLLTQTQIDLYNQLLGGISREAG
TEKIKGLNEVLNLAIQKNDETAHIIASLPHRFIPLFKQILSDRNT
LSFILEEFKSDEEVIQSFCKYKTLNRNENVLETAELFNELENSID
LTHIFISHKKLETISSALCDHWDTLRNALYERRISELTGKITKSA
KEKVQRSLKHEDINLQEIIAAGKELSEAFKQKTSEILSHAHAAAL
DQPLPTTLKKQEEKEILKSQQLDSSLGLYHLLDWFAVDESNEVDPE
FSARLTGIKLEMEPSLSFYNNKARNYATKKPYSVEKFKNFQMPTL
ASGWDVNKEKNNGAILFVKNGLYYLGIMPKQKGRYKALSFEPTK
TSEGFDKMYDYFPDAAKMIPKCSTQLKAVTAHFQTHHTTPILLSN
NFIEPLEITKEIYDLNNPEKEPKKFQTAYAKKTGDQKGYREALCK
WIDFTRDFLSKYTKTTSIDLSSLRPSSQYKDLGEYYAELNPLLYH
ISFQRIAEKEIMDAVETGKLYLFQIYNKDFAKGHHGKPNLHTLYW
TGLFSPENLAKTSIKLNGQAELFYRPKSRMKRMAHRLGEKMLNKK
LKDQKTPIPDITLYQELYDYVNHRLSHDLSDEARALLPNVITKEVS
HEIKDRRFTSDKFFFHVPITLNYQAANSPSKFNQRVNAYLKEHP
ETPIIGIDRGERNLIYITVIDSTGKILEQRSNTIQQFDYQKKLD
NREKERVAAARQAWSVVGTIKDLKQGYLSQVIHEIVDLMIHYQAVV
VLENLNFGFKSKRTGIAEKAVYQQFEKMLIDKLNCLVLKDYPAEK
VGGVLNPYQLTDQFTSFAKMGTQSGFLFYVPAPYTSKIDPLTGFV
DPFVWKTIKNHESRKHFLEGFDFLHYDVKTGDFILHFKMNRNLSF
QRGLPGFMPAWDIVFEKNETQFDAKGTPFIAGKRIVPVIENTHRFT
GRYRDLYPANELIALLEEKGIVFRDGSNILPKLLENDSDHAIDTM
VALIRSVLQMRNSNAATGEDYINSPVRDLNGVCFDSRFQNPWEPM
DADANGAYHIALKGQLLLNHLKESKDLKLQNGISNQDWLAYIQEL RN Pleckstrin
homology domain of *Homo sapiens* phospholipase C81 (hPLC81) (SEQ ID NO:
24) MDSGRDFLT LHGLQDDEDLQALLKGSQLLKVKSSSWRRERFYKLQ
EDCKTIWQESRKVMRTPESQLFSIEDIQEVRMGHRTEGLEKFARD
VPEDRCFSIVFKDQRNTLDLIAPSPADAQHWVLGLHKIIHHS GSM
DQRQKLQHWIHSCLRKADKNKDNKMSFKELQNFLKELNIQ Pleckstrin homology
domain of *Homo sapiens* Akt1 (hAkt) (SEQ ID NO: 25)
MSDVAIVKEGWLHKRGEYIKTWRPRYFLLKNDGTFIGYKERPQDV
DQREAPLNNFSVAQCQLMKTERPRPNTFIIRCLQWTTVIERTFHV
ETPEERE EWTTAIQTVADGLKKQEEEEMDFRSGSPSDNSGAEEME
VSLAKPKHRVTMNEFEYLKLLGKGTFGKVDPPV Pleckstrin homology domain of
Homo sapiens PDPK1 (hPDPK1) (SEQ ID NO: 26)
KMGPVDKRKGLFARRRQLLLTEGPHLYYVDPVNKVLKGEIPWSQE
LRPEAKNFKTFFVHTPNRTYYLMDPSGNAHKWCRKIQEVWRQRYQ SH *Homo sapiens*:
CD9 Complete Protein (SEQ ID ID NO: 27)
MSPVKGGTKCIKYL LFGFNFI FWLAGIAVLAIGLWLRFDSQTKSI
FEQETNNNNSSFYTG VYILIGAGALMMLVGFLGCCGAVQESQCML
GLFFGFLLVIFAIEIAAAIWGYSHKDEVIKEVQEFYKDTYNKLLKT
KDEPQRETLKAIHYALNCCGLAGGVEQFISDICPKKDVLETFTVK
SCPDAIKEVFDNKFHII GAVGIGIAVVMIFGMIFSMILCCAIRRN REMV *Homo sapiens*:

CD63 Complete Protein (SEQ ID NO: 28)
MAVEGGMKCVKFLLYVLLLAFCACAVGLIAVGVGAQLVLSQTHIQ
GATPGSLLPVVIIAVGVFLFLVAFVGGCGACKENYCLMITFAIFL
SLIMLVEVAAAIAGYVFRDKVMSEFNNNFRQQMENYPKNNHTASI
LDRMQADFKCCGAANYTDWEKIPSMSKNRVPDSCCINVTVGCGIN
FNEKAIHKEGECVEKIGGWLRKNVLVAAAAALGIAFVEVLGIVFAC CLVKSIRSGYEV
Homo sapiens: CD81 Complete Protein (SEQ ID NO: 29)
MGVEGCTKCIKYLFLVFNFVFWLAGGVILGVALWLRHDPQTNNLL
YLELGDKPAPNTFYVGIYILIAVGAVMMFVGFLGCGYGAIQESQCL
LGTFFTCLVILFACEVAAGIWGFVNKDQIAKDVKQFYDQALQQAV
VDDDANNAKAVVKTFFHETLDCCGSSTLTALTTSVLKNNLCPSGSN
IISNLFKEDCHQKIDDLFSGKLYLIGIAAIVVAVIMIFEMILSMV LCCGIRNSSVY *Homo sapiens*: CD47 “Self Hairpin” 10 Amino Acids (SEQ ID NO: 30) EVTELTREGE
Homo sapiens: CD47 “Self Hairpin” 21 Amino Acids (SEQ ID NO: 31)
GNYTCEVTELTREGETIIEELK *Homo sapiens*: CD47 Complete Protein (SEQ ID NO: 32) MWPLVAALLLGSACCGSAQLLFNKTKSVEFTFCNDTVVIPCFVTN
MEAQNTTEVYVKWKFKGRDIYTFDGALNKSTVPTDFSSAKIEVSQ
LLKGDASLKMDKSDAVSHTGNYTCEVTELTREGETIIEELKYRVVS
WFSPNENILIVIFPIFAILLFWGQFGIKTLKYRSGGMDEKTIALL
VAGLVITVIVIVGAILFVPGEYSLKNATGLGLIVTSTGILILLHY
YVFSTAIGLTSFVIAILVIQVIAIYILAVVGLSLCIAACIPMHGPL
LISGLSILALAQLLGLVYMKFVE Synthetic: dDZF1 (SEQ ID NO: 33)
FKCEHCRILFLDHVMFTIHMGCCHGFRDPFKCNMCGEKCDGPVGLF
VHMARNAHGEKPFYCEHCEITFRDVVMYSLHKG YHGFRDPFECNI
CGYHSQDRYEFSSHIVRGEH Synthetic: dDZF2 (SEQ ID NO: 34)
HHCQHCDMYFADNILYTIHMGCCHSCDDVFKCNMCGEKCDGPVGLF
VHMARNAHGEKPTKCVHCGIVFLDEVMYALHMSCHGFRDPFECNI
CGYHSQDRYEFSSHIVRGEH Synthetic: DmrA (SEQ ID NO: 35)
MGRGVQVETISPGDGRTFPKRGQTCVVHYTGMLEDGKKFDSSRDR
NKPFFKFMLGKQEVIRGWEEGVAQMSVGQRAKLTISPDYAYGATGH
PGIIPPHATLVFDVELLKLE Synthetic: DmrB (SEQ ID NO: 36)
MASRGVQVETISPGDGRTFPKRGQTCVVHYTGMLEDGKKVDSSRD
RNKPFFKFMLGKQEVIRGWEEGVAQMSVGQRAKLTISPDYAYGATG
HPGIIPPHATLVFDVELLKLE Synthetic: DmrC (SEQ ID NO: 37)
MGSRLWHEMWHEGLEEASRLYFGERNVKGMFEVLEPLHAMMERG
PQTLKETSFNQAYGRDLMEAQEWCRKYMKSGNVKDLLQAWDLYYH VFRRISK *Homo sapiens*/Synthetic: FKBP (SEQ ID NO: 38)
MGVQVETISPGDGRTFPKRGQTCVVHYTGMLEDGKKFDSSRDRNK
PFKFMLGKQEVIRGWEEGVAQMSVGQRAKLTISPDYAYGATGHPG
IIPPHATLVFDVELLKLE *Homo sapiens*/Synthetic: FRB (SEQ ID NO: 39)
QGMLEMWHEGLEEASRLYFGERNVKGMFEVLEPLHAMMERGPQTL
KETSFNQAYGRDLMEAQEWCRKYMKSGNVKDLLQAWDLYYHVFRR ISK Synthetic:
Anti-GCN4 scFv (SEQ ID NO: 40)
MGPDIVMTQSPSSLSASVGDRVTTITCRSSTGAVTTSNYASWVQEK
PGKLFKGLIGGTNNRAPGVPSRFSGLIGDKATLTISLQPEDFA
TYFCALWYSNHWFVGQGTKVELKRGGGGSGGGGSGGGGSSGGGSE
VKLLESGGGLVQPGGSLKLSCAVSGFSLTDYGVNWVRQAPGRGLE
WIGVIWGDGITDYN SALKDRFIISKDNGKNTVY LQMSKVRSDDTA
LYYCVTGLFDYWGGQGLTVTVSSYPYDVPDYAGGGGGSGGGGSGGG GSGGGGS
Synthetic: 10x-GCN4 Repeats (SEQ ID NO: 41)

EELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVARLKKGS
GSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVARLKK
KSGSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVA
RLKKGSGSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYHLEN
EVARLKKGSGSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYH LENEVARLKKGS

Synthetic: 24x-GCN4 Repeats (SEQ ID NO: 42)

EELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVARLKKGS
GSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVARLKK
KSGSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVA
RLKKGSGSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYHLEN
EVARLKKGSGSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYH
LENEVARLKKGSGSGEELLSKNYHLENEVARLKKGSGSGEELLSK
NYHLENEVARLKKGSGSGEELLSKNYHLENEVARLKKGSGSGEEL
LSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVARLKKGSGSG
EELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVARLKKGS
GSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVARLKK
KSGSGEELLSKNYHLENEVARLKKGSGSGEELLSKNYHLENEVA
RLKKGSGSGEELLSKNYHLENEVARLKKGSGSGEELLSKDYHLEN

EVARLKKGSGSGEELLSKNYHLENEVARLKKGS Synthetic: GFP-targeting Nanobody
(SEQ ID NO: 43) VQLVESGGALVQPGGSLRLSCAASGFPVNRYSMRWYRQAPGKERE
WVAGMSSAGDRSSYEDSVKGRFTISRDDARNTVY LQMNSLKPEDT

AVYYSNVNVGF EYWGGQTQVTVSS *Nostoc punctiforme*: Npu DnaE N-terminal
Split Intein (SEQ ID NO: 44)

CLSYETEILTVEYGLLP I G K IVEKRIECTVYSVDNNGNIYTQPVA
QWHDRGEQEVFEYCLE D G S LIRATKDHKFM TVDGQMLPIDEIFER ELDLMRVDNLPN
Nostoc punctiforme: Npu DnaE C-terminal Split Intein (SEQ ID NO: 45)

MIKIATRKYLGKQNVYDIGVERDHN FALKN GFIASNCFN Synthetic: Cfa N-Terminal
Split Intein (SEQ ID NO: 46)

CLSYDTEILTVEY G F L P I G K IVEERIECTVYTV D K N G F V Y T Q P I A
QWHNRGEQEVFEYCLE D G S IIRATKDHKFM TTDGQMLPIDEIFER GLDLKQVDGLP
Synthetic: Cfa C-Terminal Split Intein (SEQ ID NO: 47)

MVKIISRKSLGTQNVYDIGVEKDHN FLLKNGLVASN *Saccharomyces cerevisiae*: Vma N-
terminal Split Intein (SEQ ID NO: 48)

CFAKGTNVLMADGSIECIENIEVG N K V M G K D G R P R E V I K L P R G R E
TMYSVVQKSQHRAHKSDSSREVPELLKFTCNATHELWRTPRSVRR
LSRTIKGVEYFEVITFEMGQKKAPDGRIVELVKEVSKSYPISEGP
ERANELVESYRKASNKAYFEWTIEARDLSLLGSHVRKATYQTYAP ILY *Saccharomyces*
cerevisiae: Vma C-terminal Split Intein (SEQ ID NO: 49)

VLLNVLSKCAGSKKFRPAPAAAFARECRGFYFELQELKEDDY YGI
T L S D D S D H Q F L L A N Q V V V H N *Synechocystis* sp. PCC 6803: Ssp DnaE N-terminal
Split Intein (SEQ ID NO: 50)

CLSGFGTEILTVEY G P L P I G K I V S E E I N C S V Y S V D P E G R V Y T Q A I A
QWHDRGEQEVLE YELEDG S V I R A T S D H R F L T T D Y Q L L A I E E I F A R
Q L D L L T L E N I K Q T E E A L D N H R L P F P L L D A G T I K *Synechocystis* sp. PCC 6803: Ssp
DnaE C-terminal Split Intein (SEQ ID NO: 51)

MVKVIGRRSLGVQRIFDIGLPQDHN FLLANGAIAAN Synthetic: Spy Tag (SEQ ID
NO: 52) VPTIVMVDAYKRYK Synthetic: Spy Catcher (SEQ ID NO: 53)

MVTTLSGLSGEQGPSGDMTTEEDSATHIKFSKRDEDGRELAGATM
ELRDSSGKTISTWISDGHVKDFYLYPGKYTFVETAAPDGYEVATA
ITFTVNEQGQVTVNGEATKGDAHTGSSGS Bacteriophage MS2: MS2 RNA Binding

Protein (SEQ ID NO: 54)
MASNFTQFVLVDNNGGTGDVTVAPSNFANGVAEWISSNSRSQAYKV
TCSVRQSSAQNRKYTIKVEVPKVATQTVGGVELPVAAWRSYLNME
LTIPIFATNSDCELIVKAMQGLLKDGNPIPSAIAANSIGY Bacteriophage MS2: MS2
(N55K) RNA Binding Protein (SEQ ID NO: 55)
MASNFTQFVLVDNNGGTGDVTVAPSNFANGVAEWISSNSRSQAYKV
TCSVRQSSAQKRKYTIKVEVPKVATQTVGGVELPVAAWRSYLNME
LTIPIFATNSDCELIVKAMQGLLKDGNPIPSAIAANSIGY Bacteriophage MS2: MS2
(N55K)(V291) RNA Binding Protein (SEQ ID NO: 56)
MASNFTQFVLVDNNGGTGDVTVAPSNFANGIAEWISSNSRSQAYKV
TCSVRQSSAQKRKYTIKVEVPKVATQTVGGVELPVAAWRSYLNME
LTIPIFATNSDCELIVKAMQGLLKDGNPIPSAIAANSIGY Bacteriophage PP7: PP7
RNA Binding Protein (SEQ ID NO: 57)
KTIVLSVGEATRTLTEIQSTADRQIFEEKVGPLVGRLRLTASLRQ
NGAKTAYRVNLKLDQADWVDSGLPKVRYTQVWSHDVTIVANSTEA
SRKSLYDLTKSLVATSQVEDLVVNLVPLGRS Bacteriophage Mu: COM RNA Binding
Protein (SEQ ID NO: 58)
MKSIRCKNCNKLLFKADSFHIEIRCPRCKRHIIMLNACEHPTEK HCGKREKITHSDETVRY
Synthetic: Zinc Finger ZF6/10 (SEQ ID NO: 59)
STRPGERPFQCRICMRNFSIPNHLARHTRTHTGEKPFQCRICMRN
FSQSAHLKRHLRTHHTGEKPFQCRICMRNFSQDVSLVRHLKTHLRQ
KDGERPFQCRICMRNFSQAALARHTRTHTGEKPFQCRICMRNFS
QGGNLTRHLRTHHTGEKPFQCRICMRNFSQHPNLTRHLKTHLRGS Synthetic: Zinc
Finger ZF8/7 (SEQ ID NO: 60)
SRPGERPFQCRICMRNFSTMAVLRRHTRTHTGEKPFQCRICMRNF
SRREVLENHLRTHHTGEKPFQCRICMRNFSQTVNLDRLKTHLRQK
DGERPFQCRICMRNFSKKDHLHRHTRTHTGEKPFQCRICMRNFSQ
RPHLTNHLRTHHTGEKPFQCRICMRNFSVGASLKRHLKTHLRGS Synthetic: Zinc Finger
ZF9 (SEQ ID NO: 61)
SRPGERPFQCRICMRNFSDKTKLRVHTRTHTGEKPFQCRICMRNF
SVRHNLTRHLRTHHTGEKPFQCRICMRNFSQSTSLQRHLKTHLRGF Synthetic: Zinc
Finger MK10 (SEQ ID NO: 62)
SRPGERPFQCRICMRNFSRRHGLDRHTRTHTGEKPFQCRICMRNF
SDHSSLKRHLRTHHTGSQKPFQCRICMRNFSVRHNLTRHLRTHHTGE
KPFQCRICMRNFSDSNLSRHLKTHHTGSQKPFQCRICMRNFSQRS
SLVRHLRTHHTGEKPFQCRICMRNFSESGHLKRHLRTHLRGS Synthetic: FokI Zinc
Finger Nuclease 17-2 Targeting GFP (SEQ ID NO: 63)
SRPGERPFQCRICMRNFSTRQNLDTHTRTHTGEKPFQCRICMRNF
SRRDTLERHLRTHHTGEKPFQCRICMRNFSRPDALPRHLKTHLRGS
QLVKSELEEKSELRHKLKYVPHEYIELIEIARNSTQDRILEMKV
MEFFMKVYGYRGKHLGGSRKPDGAIYTVGSPIDYGVIVDTKAYSG
GYNLPIGQADEMQRYVEENQTRNKHINPNEWVKVYPSSVTEFKFL
FVSGHFKGNYKAQLTRLNHITNCNGAVLSVEELLIGGEMIKAGTL TLEEVRRKFNNGEINF
Synthetic: FokI Zinc Finger Nuclease 18-2 Targeting GFP (SEQ ID NO: 64)
SRPGERPFQCRICMRNFSSPSKLIRHTRTHTGEKPFQCRICMRNF
SDGSNLARHLRTHHTGEKPFQCRICMRNFSRVDNLPRHLKTHLRGS
QLVKSELEEKSELRHKLKYVPHEYIELIEIARNSTQDRILEMKV
MEFFMKVYGYRGKHLGGSRKPDGAIYTVGSPIDYGVIVDTKAYSG
GYNLPIGQADEMQRYVEENQTRNKHINPNEWVKVYPSSVTEFKFL
FVSGHFKGNYKAQLTRLNHITNCNGAVLSVEELLIGGEMIKAGTL TLEEVRRKFNNGEINF

Synthetic: FokI Nuclease Domain (SEQ ID NO: 65)
QLVKSELEEKKSELRHKLKYVPHEYIELIEIARNSTQDRILEMKV
MEFFMKVYGYRGKHLGGSRKPDGAIYTVGSPIDYGVIVDTKAYSG
GYNLPIGQADEMQRYVEENQTRNKHINPNEWWKVYPSSVTEFKFL
FVSGHFKGNYKAQLTRLNHITNCNGAVLSVEELLIGGEMIKAGTL TLEEVRRKFNNGEINF

Synthetic: AcuI Nuclease Domain (SEQ ID NO: 66)
VHDHKLELAKLIRNYETNRKECLNSRYNETLLRSDYLDPPFFELLG
WDIKNKAGKPTNEREVVLEEALKASASEHSKKPDYTFRLFSEKRF
FLEAKKPSVHIESDNETAKQVRRYGFTAKLKISVLSNFEYLVIYD
TSVKVDGDDTFNKARIKKYHYTEYETHFDEICDLLGRESVYSGNF
DKEWLSIENKINHFSVDTL Synthetic: Truncated AcuI Nuclease Domain (SEQ ID
NO: 67) YNETLLRSDYLDPPFFELLGWDIKNKAGKPTNEREVVLEEALKASA
SEHSKKPDYTFRLFSEKRFLEAKKPSVHIESDNETAKQVRRYG
TAKLKISVLSNFEYLVIYDTSVKVDGDDT *Ruminococcus flavefaciens*: RfxCas13d
(CasRx) (SEQ ID NO: 68)

EASIEKKKSFAKGMGVKSTLVSGSKVYMTTFAEGSDARLEKIVEG
DSIRSVNEGEAFSAEMADKNAGYKIGNAKFSHPKGYAWVANNPLY
TGPVQQDMLGLKETLEKRYFGESADGNDNICIQVIHNILDIEKIL
AEYITNAAYAVNNISGLDKDIIGFGKFSTVYTYDEFKDPEHHRAA
FNNNDKLINAIKAQYDEFDNFLDNPRLGYFGQAFFSKEGRNYIIN
YGNECYDILALLSGLRHWVHNNEEESRISRTWLYNLDKNLDNEY
ISTLNYLYDRITNELTNSFSKNSAANVNYIAETLGINPAEFAEQY
FRFSIMKEQKNLGFNITKLREVMLDRKDMSEIRKNHKVFDSIRTK
VYTMMDFVIYRYIEEDAKVAAANKSLPDNEKSLSEKDIFVINLR
GSFNDDQKDALYYDEANRIWRKLENIMHNIKEFRGNKTREYKKKD
APRLPRILPAGRDVSAFSKLMYALTMFLDGKEINDLLTTLINKFD
NIQSFLKVMPLIGVNAKFVEEYAFFKDSAKIADELRLIKSFARMG
EPIADARRAMYIDAIRILGTNLSYDELKALADTFSLDENGKNLKK
GKHGMRNFIINNVISNKRHFHYLIRYGDPAHLHEIAKNEAVVKFVL
GRIADIQKKQGQNGKNQIDRYYETCIGKDKGKSVSEKVDALTKII
TGMNYDQFDKKRSVIEDTGRENAEREKFKKIISLYLTVIYHILKN
IVNINARYVIGFHCVERDAQLYKEKGYDINLKKLEEKGFSSVTKL
CAGIDETAPDKRKDVEKEMAERAKESIDSLESANPKLYANYIKYS
DEKKAEEFTRQINREKAKTALNAYLRNTKWNVIREDLLRIDNKT
CTLFRNKAVHLEVARYVHAYINDIAEVNSYFQLYHYIMQRIIMNE
RYEKSSGKVSEYFDAVNDEKKYNDRLCLKLCVPFGYCIPRFKNLS
IEALFDRNEAAKFDKEKKKVSGNSGSG *Ruminococcus flavefaciens* & Synthetic: dead
RfxCas13d (dCasRx) (SEQ ID NO: 69)

EASIEKKKSFAKGMGVKSTLVSGSKVYMTTFAEGSDARLEKIVEG
DSIRSVNEGEAFSAEMADKNAGYKIGNAKFSHPKGYAWVANNPLY
TGPVQQDMLGLKETLEKRYFGESADGNDNICIQVIHNILDIEKIL
AEYITNAAYAVNNISGLDKDIIGFGKFSTVYTYDEFKDPEHHRAA
FNNNDKLINAIKAQYDEFDNFLDNPRLGYFGQAFFSKEGRNYIIN
YGNECYDILALLSGLAHWVVANNEEESRISRTWLYNLDKNLDNEY
ISTLNYLYDRITNELTNSFSKNSAANVNYIAETLGINPAEFAEQY
FRFSIMKEQKNLGFNITKLREVMLDRKDMSEIRKNHKVFDSIRTK
VYTMMDFVIYRYIEEDAKVAAANKSLPDNEKSLSEKDIFVINLR
GSFNDDQKDALYYDEANRIWRKLENIMHNIKEFRGNKTREYKKKD
APRLPRILPAGRDVSAFSKLMYALTMFLDGKEINDLLTTLINKFD
NIQSFLKVMPLIGVNAKFVEEYAFFKDSAKIADELRLIKSFARMG

EPIADARRADYIDALSYDELKALADTFSLDENGNNKLLK
GKHGMRNFIINNVISNKRHFHYLIRYGDPAHLHEIAKNEAVVKFVL
GRIADIQKKQGQNGKNQIDRYYETCIGKDKGKSVSEKVDALTKII
TGMNYDQFDKKRSVIEDTGRENAEREKFKKIIISLYLTVIYHILKN
IVNINARYVIGFHCVERDAQLYKEKGYDINLKKLEEKGFSSVTKL
CAGIDETAPDKRKDVEKEMAERAKESIDSLESANPKLYANYIKYS
DEKKAEEFTRQINREKAKTALNAYLRNTKWNVIREDLLRIDNKT
CTLFANKAVALVARYVHAYINDIAEVNSYFQLYHYIMQRIIMNE
RYEKSSGKVSEYFDAVNDEKKYNDRLKLLCVPFGYCIPRFKNLS
IEALFDRNEAAKFDKEKKKVSNGSGSGPKKKRKVAAAYPYDVPDY A Synthetic: L17E
(SEQ ID NO: 70) IWL TALKFLGKHA AKHEAKQQLSKL Synthetic: L17E-
Transmembrane (SEQ ID NO: 71)
IWL TALKFLGKHA AKHEAKQQLSKL NAVGQDTQE VIVVPHSLPFK
VVVISAILALVVLTIISLIILIMLWQKKPR Synthetic: KALA (SEQ ID NO: 72)
WEAKLAKALAKALAKHLAKALAKALKACEA Synthetic: KALA-Transmembrane (SEQ
ID NO: 73) WEAKLAKALAKALAKHLAKALAKALKACEANAVGQDTQE VIVVPH
SLPFKVVVISAILALVVLTIISLIILIMLWQKKPR Synthetic: Vectofusin (SEQ ID NO:
74) KKALLHAALAHLLALAHLLALLKKA Synthetic: Vectofusin-Transmembrane (SEQ
ID NO: 75) KKALLHAALAHLLALAHLLALLKKANAVGQDTQE VIVVPHSLPF
KVVVISAILALVVLTIISLIILIMLWQKKPR Synthetic: Transmembrane Domain (SEQ
ID NO: 76) NAVGQDTQE VIVVPHSLPFKVVVISAILALVVLTIISLIILIMLW QKKPR
Lactococcus lactis: Nisin A (SEQ ID NO: 77)
ITSISLCTPGCKTGALMGCMKTATCHCSIHVSK *Lactococcus lactis* NIZO 22186:
Nisin Z (SEQ ID NO: 78) ITSISLCTPGCKTGALMGCMKTATCNCSIHVSK
Lactococcus lactis subsp. *lactis* F10: Nisin F (SEQ ID NO: 79)
ITSISLCTPGCKTGALMGCMKTATCNCSVHVSK *Lactococcus lactis* 61-14: Nisin Q
(SEQ ID NO: 80) ITSISLCTPGCKTGVLMGCNLKTATCNCSVHVSK *Streptococcus*
hyointestinalis: Nisin H (SEQ ID NO: 81)
FTSISMCTPGCKTGALMTCNYKTATCHCSIKVSK *Streptococcus uberis*: Nisin U (SEQ
ID NO: 82) ITSISLCTPGCKTGILMTCPLKTATCGCHFG *Streptococcus uberis*: Nisin
U2 (SEQ ID NO: 83) VTSKSLCTPGCKTGILMTCPLKTATCGCHFG *Streptococcus*
galloyticus subsp. *pasteurianus*: Nisin P (SEQ ID NO: 84)
VTSKSLCTPGCKTGILMTCIAKTATCGCHFG *L. lactis* NZ9800: Nisin A S29A
(SEQ ID NO: 85) ITSISLCTPGCKTGALMGCMKTATCHCAIHVSK *L. lactis*
NZ9800: Nisin A S29D (SEQ ID NO: 86)
ITSISLCTPGCKTGALMGCMKTATCHCDIHVSK *L. lactis* NZ9800: Nisin A S29E
(SEQ ID NO: 87) ITSISLCTPGCKTGALMGCMKTATCHCEIHVSK *L. lactis*
NZ9800: Nisin A S29G (SEQ ID NO: 88)
ITSISLCTPGCKTGALMGCMKTATCHCGIHVSK *L. lactis* NZ9800: Nisin A K22T
(SEQ ID NO: 89) ITSISLCTPGCKTGALMGCMNTATCHCSIHVSK *L. lactis*
NZ9800: Nisin A N20P (SEQ ID NO: 90)
ITSISLCTPGCKTGALMGCPMKTATCHCSIHVSK *L. lactis* NZ9800: Nisin A M21V
(SEQ ID NO: 91) ITSISLCTPGCKTGALMGCNVKTATCHCSIHVSK *L. lactis*
NZ9800: Nisin A K22S (SEQ ID NO: 92)
ITSISLCTPGCKTGALMGCMSTATATCHCSIHVSK *L. lactis* NZ9800: Nisin Z N20K
(SEQ ID NO: 93) ITSISLCTPGCKTGALMGCKMKTATCNCSIHVSK *L. lactis*
NZ9800: Nisin Z M21K (SEQ ID NO: 94)
ITSISLCTPGCKTGALMGCNKKTATCNCSIHVSK AAV2: REP52 (SEQ ID NO: 95)
MELVGWLVDKGITSEKQWIQEDQASYISFNAASNSRSQIKAALDN
AGKIMSLTKTAPDYLVGQQPVEDISSNRIYKILELNGYDPQYAAS

VFLGWATKKFGKRTNIAEIAHTVPFYGCVNW
TNENFPFND CVDKMVIWWEEGKMTAKVVESAKAILGGSKVRVDQK
CKSSAQIDPTPVIVTSNTNMCAVIDGNSTTFEHQQPLQDRMFKFE
LTRRLDHDFGKVTQEVKDFFRWAKDHVVEVEHEFYVKKGGAKKR
PAPSDADISEPKRVRESVAQPSTSDAEASINYADRYQNKCSRHVG
MNLMLFPCRQCERMNQNSNICFTHGQKDCLECFVSESQPVSVVK
KAYQKLCYIHHIMGKVPDACTACDLVNVDLDDCIFEQ AAV2: REP78 (SEQ ID NO:
96) MPGFYEIVIKVPSDLDEHLPGISDSFVNWVAEKEWELPPDSMDL
NLIEQAPLTVAEKLQRDFLTEWRRVSKAPEALFFVQFEKGESYFH
MHVLVETTGVKSMVLGRFLSQIREKLIQRIYRGIEPTLPNWFAVT
KTRNGAGGGNKVVDECYIPNYLLPKTQPELQWAWTNMEQYLSACL
NLTERKRLVAQHLTHVSQTQEQNKENQNPNSDAPVIRSKTSARYM
ELVGWLVDKGITSEKQWIQEDQASYISFNAASNSRSQIKAALDNA
GKIMSLTKTAPDYLVGQQPVEDISSNRIYKILELNGYDPQYAASV
FLGWATKKFGKRTIWLFGPATTGKTNIAEIAHTVPFYGCVNWT
NENFPFND CVDKMVIWWEEGKMTAKVVESAKAILGGSKVRVDQKC
KSSAQIDPTPVIVTSNTNMCAVIDGNSTTFEHQQPLQDRMFKFEL
TRRLDHDFGKVTQEVKDFFRWAKDHVVEVEHEFYVKKGGAKKRP
APSDADISEPKRVRESVAQPSTSDAEASINYADRYQNKCSRHVGM
NLMLFPCRQCERMNQNSNICFTHGQKDCLECFVSESQPVSVVKK
AYQKLCYIHHIMGKVPDACTACDLVNVDLDDCIFEQ AAV2: VP1 (SEQ ID NO: 97)
MAADGYLPDWLEDTLSEGIRQWWKLKPGPPPKPAERHKDDSRGL
VLPGYKYLGPFNGLDKGEPVNEADAAALEHDKAYDRQLDSGDNPY
LKYNHADA E FQERLKEDTSFGGNLGRAVFQAKKRVLEPLGLVEEP
VKTAPGKKRPVEHSPVEPDSSSGTGKAGQQPARKRLNFGQTGDAD
SVPDPQPLGQPPAAPSGLGTNTMATGSGAPMADNNEGADGVGNSS
GNWHCDSTWMGDRVITTSTRTWALPTYNNHLYKQISSQSGASNDN
HYFGYSTPWGYFDFNRFHCHFSPRDWQRLINNNWGFRPKRLNFKL
FNIQVKEVTQNDGTTTIANNLSTVQVFTDSEYQLPYVLGSAHQG
CLPPFPADVFMVPQYGYLTLNNGSQAVGRSSFYCLEYFPSQMLRT
GNNFTFSYTFEDVPFHSSYAHSQSLDRLMNPLIDQYLYYLSRTNT
PSGTTTQSRQLQFSQAGASDIRDQSRNWLPGPCYRQQRVSKTSADN
NNSEYSWTGATKYHLNGRDSLVPNGPAMASHKDDEEKFFPQSGVL
IFGKQGSEKTNVDIEKVMITDEEEIRTTNPVATEQYGSVSTNLQR
GNRQAATADVNTQGVLPGMVWQDRDVYLQGPIWAKIPHTDGHFHP
SPLMGGFGLKHPPPQILIKNTPVPANPSTTFSAAKFASFITQYST
GQVSVEIEWELQKENS KRWNPEIQYTSNYNKS VNVDFTVDTNGVY SEPRPIGTRYLTRNL
AAV2: VP2 (SEQ ID NO: 98)
APGKKRPVEHSPVEPDSSSGTGKAGQQPARKRLNFGQTGDADSVP
DPQPLGQPPAAPSGLGTNTMATGSGAPMADNNEGADGVGNSSGNW
HCDSTWMGDRVITTSTRTWALPTYNNHLYKQISSQSGASNDNHYF
GYSTPWGYFDFNRFHCHFSPRDWQRLINNNWGFRPKRLNFKLFNI
QVKEVTQNDGTTTIANNLSTVQVFTDSEYQLPYVLGSAHQGCLP
FPADVFMVPQYGYLTLNNGSQAVGRSSFYCLEYFPSQMLRTGNN
FTFSYTFEDVPFHSSYAHSQSLDRLMNPLIDQYLYYLSRTNTPSG
TTTQSRQLQFSQAGASDIRDQSRNWLPGPCYRQQRVSKTSADNNNS
EYSWTGATKYHLNGRDSLVPNGPAMASHKDDEEKFFPQSGVLIFG
KQGSEKTNVDIEKVMITDEEEIRTTNPVATEQYGSVSTNLQRGNR
QAATADVNTQGVLPGMVWQDRDVYLQGPIWAKIPHTDGHFHP SPL
MGGFGLKHPPPQILIKNTPVPANPSTTFSAAKFASFITQYSTGQV

SVEIEWELQKENSkrWNPEIQYTSNYNksSVNVDFTVDTNGVYSEPRPIGTRYLTRNL

AAV2: VP3 (SEQ ID NO: 99)

MATGSGAPMADNNEGADGVGNSSGNWHCDSTWMGDRVITTSTRTW
ALPTYNNHLYKQISSQSGASNDNHYFGYSTPWGYFDFNRFHCHFS
PRDWQRLINNNWGFPRKRLNFKLFNIQVKEVTQNDGTTTIANNL
STVQVFTDSEYQLPYVLGSAHQGCLPPFPADVFMVPQYGYLTLNN
GSQAVGRSSFYCLEYFPSQMLRTGNNFTFSYTFEDVPFHSSYAHS
QSLDRLMNPLIDQYLYLSRTNTPSGTTTQSRLQFSQAGASDIRD
QSRNWLPGPCYRQQRVSKTSADNNNSEYSWTGATKYHLNGRDSL
NPGPAMASHKDDEEKFFPQSGVLIFGKQGSEKTNVDIEKVMITDE
EEIRTTNPVATEQYGSVSTNLQRGNRQAATADVNTQGVLPGMVWQ
DRDVYLQGPiWAKIPHTDGHFHPSPLMGGFGLKHPPPQILIKNTP
VPANPSTTFSAAKFASFITQYSTGQVSVEIEWELQKENSkrWNPE

IQYTSNYNksSVNVDFTVDTNGVYSEPRPIGTRYLTRNL Relevant RNA Sequences

(5'-3') Synthetic: MS2 Stem Loop spCas9 Scaffold RNA for sgRNA with Terminator Example 1 (SEQ ID NO: 100)

GUUUUAGAGCUAGGCCAACAUAGGAUCACCCAUGUCUGCAGGGC
CUAGCAAGUUAUUUUUAAGGCUAGUCCGUUAUCAACUUGGCCAAC
UGAGGAUCACCCAUGUCUGCAGGGCCAAGUGGCACCGAGUCGGUG CUUUUUUU

Synthetic: MS2 Stem Loop spCas9 Scaffold RNA for sgRNA with Terminator Example 2 (SEQ ID NO: 101)

GUUUUAGAGCUAGGCCAACAUAGGAUCACCCAUGUCUGCAGGGC
CUAGCAAGUUAUUUUUAAGGCUAGUCCGUUAUCAACUUGGCCAAC
UGAGGAUCACCCAUGUCUGCAGGGCCAAGUGGCACCGAGUCGGUG
CGGGAGCACAUGAGGAUCACCCAUGUGCGACUCCACAGUCACUG

GGGAGUCUUCUUUUUUUU Synthetic: MS2 Stem Loop spCas9 Scaffold RNA for sgRNA with Terminator Example 3 (SEQ ID NO: 102)

GUUUAAAGAGCUAUGCUGGAAACAGCAUAGCAAGUUAUUUUUAAGGC
UAGUCCGUUAUCAACUUGAAAAAGUGGCACCGAGUCGGUGCGGGA
GCACAUGAGGAUCACCCAUGUGCGACUCCACAGUCACUGGGGAG

UCUUCUUUUUUUU Synthetic: 4xMS2 Stem Loop RNA Scaffold Example (SEQ ID NO: 103)

UUCUAGAUAUCAUGGAAACAUGAGGAUCACCCAUAUCUGCAGUCGAC
AUCGAAACAUGAGGAUCACCCAUGUCUGCAGUCGACAUCGAAACA
UGAGGAUCACCCAUGUCUGCAGUCGACAUCGAAACAUGAGGAUCA
CCAUGUCUGCAGUCGACAUCGAAAUCAUAAGCUUCAGAUCAUCCUAG

Synthetic: MS2 Stem Loop Example 1 (SEQ ID NO: 104)

ACAUGAGGAUCACCCAUGU Synthetic: MS2 Stem Loop Example 2 (SEQ ID NO: 105)

ACAUGAGGAUCACCCAUAU Synthetic: MS2 Stem Loop Example 3

(SEQ ID NO: 106) CCACAGUCACUGGG Synthetic: 2xMS2 Stem Loop Example (SEQ ID NO: 107)

ACAUGAGGAUCACCCAUGUCUGCAGGGCCUAGCAAGUUAUUUUUA

GGCUAGUCCGUUAUCAACUUGGCCAACAUAGAGGAUCACCCAUGU Synthetic:

2xPP7 Stem Loop spCas9 Scaffold RNA for sgRNA with Terminator Example (SEQ ID NO: 108)

GUUUUAGAGCUAGGCCGGAGCAGACGAUAUGGCGUCGCUCCGGCC
UAGCAAGUUAUUUUUAAGGCUAGUCCGUUAUCAACUUGGCCGGAGC

AGACGAUAUGGCGUCGCUCCGGCCAAGUGGCACCGAGUCGGUGCU UUUUUU

Synthetic: PP7 Stem Loop Example (SEQ ID NO: 109)

GCCGGAGCAGACGAUAUGGCGUCGCUCCGGCC Synthetic: COM Stem Loop

spCas9 Scaffold RNA for sgRNA with Terminator Example (SEQ ID NO: 110)

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OTHER EMBODIMENTS

(209) It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

Claims

1. A cell comprising: (a) an exogenous nucleic acid molecule that encodes a human endogenous retrovirus (HERV) glycoprotein, and (b) an exogenous nucleic acid molecule that encodes a protein comprising a therapeutic cargo fused to a plasma membrane recruitment domain, wherein the plasma membrane recruitment domain is not a viral GAG protein or a CD63 protein.
2. The cell of claim 1, wherein the therapeutic cargo comprises a DNA binding protein or an RNA binding protein.
3. The cell of claim 1, wherein the therapeutic cargo comprises a gene editing reagent.
4. The cell of claim 3, wherein the gene editing reagent comprises a zinc finger (ZF), a transcription activator-like effector (TALE), a CRISPR-based genome editing or modulating protein; or a ribonucleoprotein complex (RNP) comprising the CRISPR-based genome editing or modulating protein.
5. The cell of claim 3, wherein the gene editing reagent comprises an RNA guided protein.
6. The cell of claim 3, wherein the gene editing reagent is configured to bind a guide RNA.
7. The cell of claim 1, wherein the plasma membrane recruitment domain is a human endogenous gag protein.
8. The cell of claim 7, wherein the human endogenous gag protein comprises a HERV GAG protein.
9. The cell of claim 7, wherein the human endogenous gag protein comprises a HERV GAG consensus sequence.
10. The cell of claim 1, wherein the plasma membrane recruitment domain is a human plasma membrane recruitment domain.
11. The cell of claim 1, wherein the plasma membrane recruitment domain comprises a pleckstrin homology (PH) domain.
12. The cell of claim 11, wherein the PH domain is selected from the group consisting of a pleckstrin homology domain of phospholipase C δ 1 (PLC δ 1), Pleckstrin homology domain of Akt1 or a mutant thereof, and Pleckstrin homology domain of PDK1.
13. The cell of claim 11, wherein the PH domain is a mutant pleckstrin homology domain of Akt1 that comprises an amino acid substitution relative to a corresponding wild type Pleckstrin homology domain of Akt1.
14. The cell of claim 13, wherein the amino acid substitution comprises an amino acid substitution of amino acid residue E17 relative to the corresponding wild type pleckstrin homology domain of Akt1.
15. The cell of claim 14, wherein the amino acid substitution is E17K.
16. The cell of claim 1, wherein the plasma membrane recruitment domain comprises the amino acid sequence selected from the group consisting of the amino acid sequence of any one of SEQ ID NO: 19, 24-27, and 29-32.
17. The cell of claim 1, wherein the HERV glycoprotein is selected from the group consisting of:

hENVH1, hENVH2, hENVH3, hENVK1, hENVK2, hENVK3, hENVK4, hENVK5, hENVK6, hENVT, hENVW, hENVFRD, hENVR, hENVR(b), hENVF(c)2, hENVF(c)1, and hENVKcon.

18. The cell of claim 1, wherein the HERV glycoprotein comprises the amino acid sequence of any one of SEQ ID NOs: 2-18.

19. The cell of claim 1, wherein the HERV glycoprotein is fused to a targeting polypeptide.

20. The cell of claim 19, wherein the targeting polypeptide comprises a single-chain variable fragment (ScFv).

21. The cell of claim 1, wherein the cell further comprises an exogenous nucleic acid molecule encoding HERV GAG protein.

22. The cell of claim 21, wherein the HERV GAG protein comprises a gag protein sequence of HERV-K113, HERV-K101, HERV-K102, HERV-K104, HERV-K107, HERV-K108, HERV-K109, HERV-K115, HERV-K11p22, or HERV-K12q13.

23. The cell of claim 22, wherein the HERV GAG protein comprises a HERV GAG consensus sequence.

24. The cell of claim 23, wherein the HERV GAG consensus sequence comprises the amino acid sequence of SEQ ID NO: 19.

25. A population of cells comprising the cell of claim 1.

26. A composition comprising: a nucleic acid molecule that encodes a HERV glycoprotein, and a nucleic acid molecule that encodes a protein comprising a therapeutic cargo fused to a plasma membrane recruitment domain, wherein the plasma membrane recruitment domain is not a viral GAG protein or a CD63 protein.

27. A method of producing a particle that comprises a therapeutic cargo, the method comprising: providing a cell comprising: (a) an exogenous nucleic acid molecule that encodes a HERV glycoprotein, and (b) an exogenous nucleic acid molecule that encodes a protein comprising a therapeutic cargo fused to a plasma membrane recruitment domain, wherein the plasma membrane recruitment domain is not a viral gag protein or a CD63 protein, and maintaining the cell under conditions such that the cell produces the particle.

28. The method of claim 27, wherein the method further comprises purifying the particle from the cell.
